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DEPARTMENT OF DEFENSE HANDBOOK

DIGITAL ENGINEERING AND MODELING PRACTICES

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FOREWORD

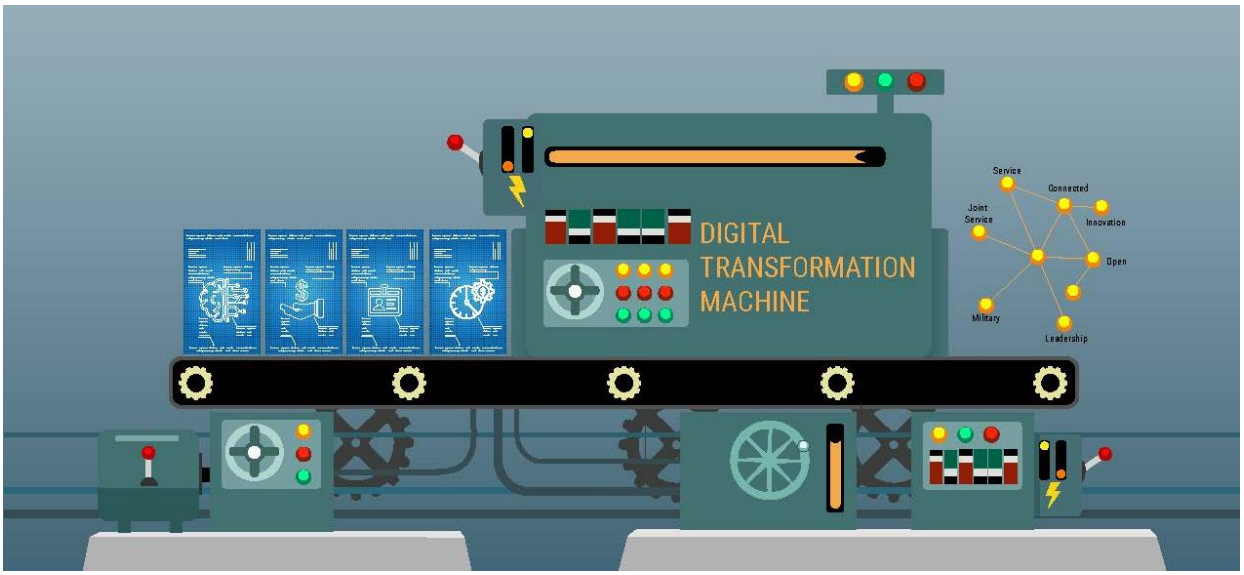
1. This handbook is approved for use by all Departments and Agencies of the Department of Defense (DoD).

2. This handbook will be developed using an iterative process as follows:

a. Initial release – provides guidance on current practices with terms, definitions, and descriptions to meet the scope and objective of this document.

b. Iterative sprints – includes additional updates to the content as the digital engineering (DE) and modeling approach matures, which will be reflected in future revisions to this handbook.

3. Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard, DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.



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1. SCOPE

1.1 Scope. This handbook serves the DoD community, including DoD support contractors, to help identify how digital engineering (DE) and modeling activities can be considered for all disciplines and functional areas within an acquisition lifecycle. This handbook is for guidance only and cannot be cited as a requirement.

1.2 Handbook purpose. The purpose of this handbook is to provide guidance on DE and modeling strategies to achieve a high level of confidence in the practices of integration and collaboration technologies. This also includes maximizing agility, interoperability, reusability, and scalability across the DoD by shifting from traditional document-centric activities to data-centric activities. This shift will allow data-driven decisions and continuous access to authoritative data by the DoD community, ensuring the right information is delivered to the right person at the right time throughout the lifecycle.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed below are not necessarily all of the documents referenced herein, but are those needed to understand the information provided by this handbook.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-963 - Data Item Descriptions (DIDs)

MIL-STD-31000 - Technical Data Packages

DEPARTMENT OF DEFENSE HANDBOOKS

MIL-HDBK-61 - Configuration Management Guidance

(Copies of these documents are available online at https://quicksearch.dla.mil/.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein.

DEPARTMENT OF DEFENSE DIRECTIVES

DoDD 5000.01 - The Defense Acquisition System

DEPARTMENT OF DEFENSE INSTRUCTIONS

DoDI 5000.02 - Operation of the Adaptive Acquisition Framework

DoDI 5000.75 - Business Systems Requirements and Acquisition

DoDI 5000.86 - Acquisition Intelligence

DoDI 5000.88 - Engineering of Defense Systems

DoDI 5000.91 - Product Support Management for the Adaptive Acquisition Framework

DoDI 5000.93 - Use of Additive Manufacturing in the DoD

DoDI 8500.01 - Cybersecurity

DEPARTMENT OF DEFENSE MANUALS

DoDM 4120.24 - Defense Standardization Program (DSP) Procedures

(Copies of these documents are available online at https://www.acq.osd.mil/se/.)

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DEPARTMENT OF DEFENSE PUBLICATIONS

Department of Defense Digital Engineering Strategy

SEP - Systems Engineering Plan (SEP) Outline, Version 4.0

(Copies of these documents are available online at https://ac.cto.mil/erpo/.)

Department of Defense Architecture Framework

(Copies of this document are available online at https://dodcio.defense.gov/library/dod-architecture-framework/.)

DoD PM's Guidebook for Integrating the Cybersecurity Risk Management Framework (RMF) into the System Acquisition Lifecycle

(Copies of this document are available online at https://www.dau.edu.)

Mission Engineering Guide

(Copies of this document are available online at https://aaf.dau.edu/aaf/policies/.)

UNITED STATES AIR FORCE PUBLICATIONS

Guidance for e-Program Designations

(Copies of this document are available online at https://software.af.mil/.)

Air Force Digital Guide

(Copies of this document are available online with an account at https://wss.apan.org/af/aflcmc/.)

UNITED STATES NAVY PUBLICATIONS

US Naval & Marine Corps Digital Systems Engineering Transformation Strategy

(Copies of this document are available online with an account at https://portal.lift.mhpcc.hpc.mil/.)

UNITED STATES SPACE FORCE PUBLICATIONS

USSF Vision for a Digital Service

(Copies of this document are available online at https://media.defense.gov.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME Y14.5 - Dimensioning and Tolerancing

ASME Y14.34 - Associated Lists

ASME Y14.41 - Digital Product Definition Data Practices

ASME Y14.47 - Model Organization Practices

ASME Y14.100 - Engineering Drawing Practices

(Copies of these documents are available online at www.asme.org.)

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO) PUBLICATIONS

ISO 10303 - Automation Systems and Integration – Product Data Representation and Exchange

ISO/IEC 12207 - Software Lifecycle Processes

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ISO 32000-1 - Portable Document Format

(Copies of these documents are available online at www.iso.org.)

NATIONAL SHIPBUILDING RESEARCH PROGRAM (NSRP) ADVANCED SHIPBUILDING ENTERPRISE (ASE)

NSRP ASE 2009-350 - 2008 Systems Technology Panel Project: Interim SCIM and STEP Implementation

(Copies of this document are available online at www.nsrp.org.)

OBJECT MANAGEMENT GROUP (OMG)

OMG Systems Modeling Language (SysML™)

OMG® Unified Modeling Language® (UML®)

(Copies of these documents are available online at https://www.omg.org/spec/.)

SAE INTERNATIONAL

SAE AS-6969 - Data Dictionary used for Quantities Used in Cyber-Physical System

SAE EIA-649 - Configuration Management Standard

SAE EIA-649-1 - Configuration Management Requirements for Defense Contracts

(Copies of these documents are available online at www.sae.org.)

3. ACRONYMS AND DEFINITIONS

3.1 Acronyms.

Acronym	**Term**
2D	2-Dimensional
3D	3-Dimensional
3Di	3-Dimensional Intelligent
AP	Application Protocol
BOM	Bill of Materials
CAD	Computer-Aided Design
CAGE	Commercial and Government Entity
CCB	Configuration Control Board
CDRL	Contract Data Requirements List
CM	Configuration Management
COTS	Commercial Off the Shelf
DE	Digital Engineering
DEE	Digital Engineering Ecosystem
DID	Data Item Description
DISR	Defense Information Technology Standards Registry
DLA	Defense Logistics Agency
DoD	Department of Defense
DoDAF	Department of Defense Architecture Framework

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DSM	Digital System Model
FACE	Future Airborne Capability Environment
GRA	Government Reference Architecture
GRM	Government Reference Model
HOST	Hardware Open Systems Technologies
HPCMP	High Performance Computing Modernization Program
HTML	Hypertext Markup Language
IaaS	Infrastructure as a Service
IDE	Integrated Digital Environment
IEEE	Institute of Electrical and Electronics Engineers
IGES	Initial Graphics Exchange Specification
IP	Intellectual Property
IT	Information Technology
JAMMEX	Joint Additive Manufactured Model Exchange
JT	Jupiter Tessellation
MBD	Model-Based Definition
MBDes/EA	Model-Based Design and Engineering Analysis
MBE	Model-Based Enterprise
MBI/QA	Model-Based Inspection/Quality Assurance
MBM	Model-Based Manufacturing
MBME	Model-Based Mission Engineering
MBPS	Model-Based Product Support
MBReq	Model-Based Requirements
MBS	Model-Based Sustainment
MBSD	Model-Based System Development
MBSE	Model-Based Systems Engineering
MBT&E	Model-Based Testing and Evaluation
MOSA	Modular Open Systems Approach
MRO	Maintenance, Repair, and Operations
MT	Mission Threads
NAVAIR	Naval Air Systems Command
NAVSEA	Naval Sea Systems Command
NOBLE	Navy Operational Business Logistics Enterprise
NMMES-TR	Navy Maritime Maintenance Enterprise Solution – Tech Refresh
O&S	Operations and Support
OMS	Open Mission Systems
OV	Operational Viewpoint
PaaS	Platform as a Service
PDF	Portable Document Format
PDM	Product Data Management
PLM	Product Lifecycle Management

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PM	Program Manager
PTC	Parametric Technology Corporation
RFP	Request for Proposal
RMF	Risk Management Framework
RVTM	Requirement Verification Traceability Matrix
SCM	Supply Chain Management
SEP	Systems Engineering Plan
SoS	System of Systems
SOSA	Sensor Open Systems Architecture
STEP	Standard for the Exchange of Product Model Data
StdV	Standards Viewpoint
SvcV	Services Viewpoint
SysML™	Systems Modeling Language
TDP	Technical Data Package
TDPL	Technical Data Package List
TTP	Tactics, Techniques, and Procedures
TWH	Technical Warrant Holder
UML®	Unified Modeling Language
USMC	United States Marine Corps
UxS	Unmanned Systems
XMI	XML Metadata Exchange
XML	Extensible Markup Language

3.2 Definitions. Master terms and definitions are as specified in DoDD 5000.01. For the purposes of this handbook, the following definitions apply:

3.2.1 3-Dimensional intelligent (3Di) technical data. A 3-dimensional (3D) viewable computer-aided design (CAD) representation that details the complete technical description of the required design configuration of an item provided in a widely available software format (e.g., ISO 32000-1 portable document format [PDF]), as specified in MIL-STD-31000.

3.2.2 3D model. A digital representation describing the form of a physical object, displayed via specialized software.

3.2.3 3Di format. The standard arrangement and organization of information within a 3Di viewable representation of an item. This includes features such as the size and arrangement of information blocks (e.g., title blocks), notes, lists, revision information, view states, restriction notices, and the use of optional or supplemental blocks, as specified in MIL-STD-31000.

3.2.4 Annotation. Visible dimensions, tolerances, notes, text, or symbols, as specified in ASME Y14.41 or ASME Y14.5.

3.2.5 Architecture. The structure of components, their relationships, and the principles and guidelines governing their design and evolution over time.

3.2.6 Architecture data. Facts, characteristics, and concepts that define the structure of a system and the interrelationships between its parts and its environment. The data is created in a manner suitable for communication, interpretation, or processing by humans or by automatic means. For complex systems, a framework

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of conventions, principles, and practices is used for organizing and presenting the architecture data within a specific domain of application or community of stakeholders.

3.2.7 Architecture framework. An architecture framework establishes a common practice for creating, interpreting, analyzing, and using architecture descriptions within a particular domain of application or stakeholder community. Examples of architecture frameworks include ministry of defence architecture framework (MODAF), The Open Group Architecture Framework, Kruchten's 4+1 view model, and reference model of open distributed processing.

3.2.8 Attribute. A dimension, tolerance, note, text, or symbol required to complete the product definition or feature of the product that is not visible, but available upon interrogation of the annotated model, as specified in ASME Y14.41 or ASME Y14.5.

3.2.9 Attribute service. A service that provides a consumer of data with a common access point to authoritative data obtained from one or more sources that are considered the originating authoritative source.

3.2.10 Authoritative. Able to be trusted as being accurate or true; reliable.

3.2.11 Authoritative automation. The technique, method, or system of operating or controlling a trusted and reliable process by highly automatic means, as by electronic devices, reducing human intervention to a minimum.

3.2.12 Authoritative source. A managed repository of valid or trusted data that is recognized by an appropriate set of governance entities and supports the governance entity's business environment because its authority or authenticity is widely recognized by experts in the field.

3.2.13 Authoritative source of truth. An entity, such as a person, governing body, or system, that applies expert judgement and rules to proclaim an artifact is valid and originates from a legitimate source.

3.2.14 Authoritative visualization. A trusted and reliable virtual version of something, including virtual computer hardware platforms, storage devices, and computer network resources.

3.2.15 Automation. The technique, method, or system of operating or controlling a process by highly automatic means, as by electronic devices, reducing human intervention to a minimum.

3.2.16 Bill of materials (BOM). A listing of the parts, components, and sub-assemblies; and the quantities of each; needed to define, manufacture, or support an end product. BOMs typically identify by nomenclature and product identifiers.

3.2.17 BOM, structured. A listing of components, typically identified by nomenclature and product identifier(s), that make up a product, arranged in a hierarchy such that product structure and component quantities can be determined for the end item itself and for all assemblies and subassemblies contained in the end item. The structured BOM may consist of three subtypes as defined below:

3.2.17.1 Engineering BOM (E-BOM). A structured BOM that defines a product.

3.2.17.2 Manufacturing BOM (M-BOM). A structured BOM that identifies components and materials necessary to manufacture a product.

3.2.17.3 Provisioning BOM (P-BOM). A structured BOM that identifies components necessary to logistically support a product by identification of spare and repair components.

3.2.18 Commercial engineering design data. Engineering design data prepared by a commercial design activity, in accordance with that activity's documentation standards and practices, to support the development and manufacture of a commercially developed product, as specified in MIL-STD-31000.

3.2.19 Company standard. A company document that establishes engineering and technical limitations and applications for items, materials, processes, methods, designs, and engineering practices unique to that company.

3.2.20 Computer software. Computer programs, source code, source code listings, object code listings, design details, algorithms, processes, flow charts, formulae, and related material that would enable the software to be

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reproduced, recreated, or recompiled. Computer software does not include computer databases or computer software documentation, as specified in DFARS Clause 252.227-7014.

3.2.21 Computer software documentation. Owner's manuals, user's manuals, installation instructions, operating instructions, and other similar items, regardless of storage medium, that explain the capabilities of the computer software or provide instructions for using the software, as specified in DFARS Clause 252.227-7014. ISO/IEC Standard 12207 uses the term "software lifecycle data" to address software documentation.

3.2.22 Conceptual data model. A map of concepts and their relationships. This describes the semantics of an organization and represents a series of assertions about its nature. Specifically, it describes the things of significance to an organization (entity classes), about which it is inclined to collect information, and its characteristics (attributes) and the associations between pairs of those things of significance (relationships).

3.2.23 Contract data requirements list (CDRL). A list of authorized data requirements for a specific procurement that forms part of a contract. It is comprised of a single DD Form 1423, or a series of DD Form 1423s, containing data requirements and delivery information. The CDRL is the standard delivery format for identifying potential data requirements in a solicitation and deliverable data requirements in a contract and identifies what data products are to be formally delivered to the Government by a contractor, as well as when and possibly how (e.g., format and quantity) they are to be delivered.

3.2.24 Data access. A user's ability to effectively and securely create, manipulate, deposit or upload, retrieve or download, store, and exchange all data residing within a structured electronic collaborative database or other repository, as specified in MIL-HDBK-61. Data access is not interchangeable with Data Delivery

3.2.25 Data delivery. The physical delivery or transmission of information and files from one source or ownership to another.

3.2.26 Data model. The definition of all data elements (and their relationships) to be maintained within a product data management (PDM) system. The data model defines the database schema for the PDM system.

3.2.27 DevSecOps. A set of software development practices that combines software development (Dev), security (Sec), and information technology operations (Ops) to secure the outcome and shorten the development lifecycle. DevSecOps supports a Government reference architecture (GRA), or Government-owned reference architecture definition.

3.2.28 Digital artifact. Products (e.g., documents, diagrams, presentations or briefings, charts, spreadsheets, processes, or workflows) within, or generated from, the digital engineering ecosystem (DEE) that provide ways or means to visualize, communicate, or deliver data, information, and knowledge to stakeholders.

3.2.29 Digital engineering (DE). An integrated, computation-based approach that uses authoritative sources of system data and models as a continuum across disciplines to support lifecycle activities.

3.2.30 Digital engineering ecosystem (DEE). The interconnected infrastructure, environment, and methodology (processes, methods, and tools) used to store, access, analyze, and visualize evolving systems' data and models to address the needs of stakeholders.

3.2.31 Digital system model. A digital representation of a defense system, generated by all stakeholders, that integrates the authoritative technical data and associated artifacts which define all aspects of the system for the specific activities throughout the system lifecycle.

3.2.32 Digital thread. A communication framework that connects a system's data across functional perspectives.

3.2.33 Digital twin. A virtual replica of a physical entity that is synchronized across time. Digital twins exist to replicate configuration, performance, or history of a system. Two primary sub-categories of digital twin are as follows.

3.2.33.1 Digital instance. A virtual replica of the physical configuration of an existing entity. The digital instance typically exists to replicate each individual configuration of a product as-built or as-maintained.

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3.2.33.2 Digital prototype. An integrated multi-physics, multiscale, probabilistic model of a system design. The digital prototype may use sensor information and input data to simulate the performance of its corresponding physical twin. The digital prototype may exist prior to realization of its physical counterpart.

3.2.34 Engineering design data. Engineering drawings; 3Di viewable, native CAD models; neutral CAD models; or a combination of these, that define an item by means of graphic and textual presentations, the physical and functional requirements of an item, sufficient to fulfill its technical data package (TDP) element and level requirements, as specified in MIL-STD-31000. Engineering design data comes in three levels corresponding to its TDP level requirements as defined in the following paragraphs.

3.2.34.1 Conceptual design data. Engineering design data that describes the engineering concepts on which a proposed technology or design approach is based.

3.2.34.2 Developmental design data. Engineering design data that describes the physical and performance characteristics of a specific design approach to the extent necessary to permit the analytical evaluation of its ability to meet specified requirements and enable the development, manufacture, and testing of prototype or experimental materiel.

3.2.34.3 Product design data. Engineering design data that describes the complete physical and performance characteristics of an item or component in sufficient detail to ensure that an item or component produced in accordance with this data will be essentially identical to the original item or component.

3.2.35 Product data management (PDM). The business function, often within product lifecycle management (PLM), that is responsible for the management and publication of product data.

3.2.36 Fully annotated model. A 3D CAD dataset in which all necessary features to fully define the item (full design disclosure including dimensions, tolerances, materials, notes, surface finishes, etc.) are included in a readily viewable form, as specified in MIL-STD-31000, ASME Y14.41, ASME Y14.41.1, or ASME Y14.5.

3.2.37 Government reference architecture (GRA). The reference architecture provided by the Government to guide and constrain the system design, development, production, and sustainment processes.

3.2.38 Government reference model (GRM). Provides the architecture views used to support business case development and investment decisions, monitor capability development, and service improvements. The GRM can be provided to potential bidders for contracted efforts to further acquisition, development, and sustainment.

3.2.39 Integrated digital environment (IDE). A compilation of data, models, and tools for collaboration, analysis, and visualization across all functional domains. IDE includes the methodology and specification for data, models, and tools arrangement with processes and procedures to exploit informational results.

3.2.40 Interoperability. The ability to exchange information and operate effectively together.

3.2.41 Jupiter tessellation (JT) file. A 3D data format used for product visualization, collaboration, CAD data exchange, and long-term data retention. It contains a combination of 3D geometry, product and manufacturing information, and metadata that is a compression of a native CAD file.

3.2.42 Limited design disclosure model. A 3D CAD dataset that is sufficiently defined to provide a visual understanding of the item, but does not contain full design disclosure. Generally, interface characteristics and features such as weight and center of gravity will be sufficiently defined for the intended purpose. Sometimes referred to as an envelope model, shrink-wrap model, or cosmetic model, as specified in MIL-STD-31000.

3.2.43 Logical data model. A data model of a specific problem domain expressed independently of a particular database management product or storage technology (physical data model) but in terms of data structures such as relational tables and columns, object-oriented classes, or extensible markup language (XML) tags. This is as opposed to a conceptual data model, which describes the semantics of an organization without reference to technology.

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3.2.44 Metadata. Generally defined to be data about data. For the purpose of this handbook, metadata is defined to be data about design data and its defining engineering design data documents. Metadata is used by the procuring activity to store, manage, and provide access to TDP elements.

3.2.45 Model. A version of a concept, relationship, and structure of a system.

3.2.46 Modeling and simulation. As a noun, the discipline that comprises the development and use of models and simulations. As a verb, the use of models and simulations, either statically or over time, to develop data as a basis for making managerial or technical decisions.

3.2.47 Model-based definition (MBD). A dataset containing the exact solid representation of a product, its associated 3D geometry, and the 3D annotation of its dimensions, tolerances, materials, finishes, and other notes to specify a complete product definition. Additional definitions are as specified in ASME Y14.41.1, ASME Y14.47, or ASME Y14.5.

3.2.48 Model-based engineering. An approach to engineering that uses models as an integral part of the design process that includes the requirements, analysis, design, implementation, and verification of a capability, system, and product throughout the acquisition lifecycle.

3.2.49 Model-based enterprise (MBE). An organization that uses MBD, as well as other digital models and associated datasets, in an integrated and collaborative environment, shared across the enterprise, as the foundation of the product lifecycle.

3.2.50 Model-based manufacturing (MBM). The use of an MBD and other analytical models to design, maintain, and execute the manufacturing process of an item.

3.2.51 Model-based product support (MBPS). A digital transformation program that modernizes the Navy's current maritime logistics data systems and that provides configuration management, provisioning, readiness modeling, and technical data management support for ships/subs and weapon systems. It consolidates the current disparate logistics information technology (IT) systems, institutionalizes naval enterprise collaboration, and provides modeling and simulation capabilities.

3.2.52 Model-based requirements (MBReq). An approach to MBE that uses modeling to outline the DoD's goals and success conditions for the system.

3.2.53 Model-based software development (MBSD). An approach to MBE that uses graphical notations and models to design necessary software systems.

3.2.54 Model-based sustainment (MBS). The use of an MBD and other analytical models to design, maintain, and execute the sustainment and support strategy of an end item.

3.2.55 Model-based systems engineering (MBSE). The formalized application of modeling to support system requirements, design, analysis, verification, and validation activities beginning in the conceptual design phase and continuing throughout development and later lifecycle phases.

3.2.56 Model-based testing (MBT). An approach to MBE that uses models to generate test cases and support the verification and validation of systems.

3.2.57 Model types:

3.2.57.1 Analytical model. A method of mathematical modeling that frames a process with many parameters and calculates an exact solution without computer simulation.

3.2.57.2 Mathematical model. A symbolic model whose properties are expressed in mathematical symbols and relationships.

3.2.57.3 Numerical model. A method of mathematical modeling that runs an initial condition with varying parameters and approximates a solution.

3.2.57.4 Physical data model. A representation of a data design as implemented, or intended to be implemented, in a database management system. In the lifecycle of a project, it typically derives from a logical data

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model, though it may be reverse engineered from a given database implementation. A complete physical data model will include all the database artifacts required to create relationships between tables or to achieve performance goals, such as indexes, constraint definitions, linking tables, partitioned tables, or clusters. Analysts can usually use a physical data model to calculate storage estimates; it may include specific storage allocation details for a given database system.

3.2.57.5 Physical model. A mockup that represents an actual system.

3.2.57.6 Simulation model. A model that emulates a real-life system through computer software.

3.2.58 Modular open systems approach (MOSA). MOSA is an integrated business and technical strategy to achieve competitive and affordable acquisition and sustainment over the system life cycle.

3.2.59 Product lifecycle management (PLM). A strategic business approach that applies a consistent set of business solutions in support of the collaborative creation, management, dissemination, and use of product definition information (product data) across the extended enterprise, and spanning from product concept to end of life, integrating people, processes, business systems, and information.

3.2.60 Reference architecture. Authoritative sources of data about a specific subject area that guides and constrains the instantiations of multiple architectures and solutions.

3.2.61 Technical data. Recorded information, regardless of the form or method of the recording, of a scientific or technical nature (including computer software documentation). The term does not include computer software or data incidental to contract administration, such as financial or management information, as specified in DFARS Clause 252.227-7013.

3.2.62 Technical data - native CAD data. CAD data as created in its original authoring software format. In general, only the original authoring software format is capable of reading, editing, and interpreting native CAD data. Examples of native CAD data authoring software include CREO®, CATIA®, Autodesk®, and SolidWorks®, or equal. Generally, native CAD data is the only format suitable to serve as master technical data. Native CAD data may also serve as authoritative technical data. This definition is as specified in MIL-STD-31000.

3.2.63 Technical data - neutral CAD data. CAD data that is derived from the native format and converted into a format that can be imported into other CAD software. Neutral CAD data is created to a widely available national or international standard (e.g., standard for the exchange of product model data [STEP], initial graphics exchange specification [IGES], etc.). In general, neutral CAD data cannot serve as master technical data but may serve as either reference or authoritative technical data. This definition is as specified in MIL-STD-31000.

3.2.64 Technical data - viewable CAD data. CAD data that is derived from the native format and converted into a format that can be displayed by widely available software and is for purposes of defining design intent in a human readable format (see Appendix H). In general, viewable CAD data cannot serve as master technical data but may serve as either reference or authoritative technical data. This definition is as specified in MIL-STD-31000.

3.2.65 Technical data origin - master technical data. A set of technical data that is the controlling source for any subsequent technical data output. All changes to the technical data originates on the master technical dataset. There can be only one instance of the master technical data, which is typically in the native CAD format and maintained by the current design activity. This definition is as specified in MIL-STD-31000.

3.2.66 Technical data origin - derivative technical data. A set of technical data generated on master technical data. Typically, derivative technical data is created to change its format to suit a different purpose; for example, converting master, native format data to neutral or viewable data (e.g., STEP, IGES, PDF) in order for it to be read by a wider range of software applications. Changes to derivative technical data originate on the master technical dataset. This definition is as specified in MIL-STD-31000.

3.2.67 Technical data package (TDP). The authoritative technical description of an item. This technical description supports the acquisition, production, inspection, engineering, and logistics support of the item. The description defines the required design configuration and performance requirements, as well as procedures required to ensure adequacy of item performance. It consists of applicable technical data such as models, engineering design

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data, associated lists, specifications, standards, performance requirements, quality assurance provisions, software documentation, and packaging details. This definition is as specified in MIL-STD-31000.

3.2.68 Technical data package list (TDPL). An index of all documents contained in a TDP established at a point in time. TDPLs contain a list of all specifications, engineering design data (e.g., drawings, models, 3Di viewables, etc.), quality assurance provisions , inspection documents, packaging documents or other documents which comprise the TDP listed by nomenclature, document commercial and Government entity (CAGE) code, document number, distribution code, revision, and date. This definition is as specified in MIL-STD-31000.

3.2.69 Validation status - authoritative technical data. A set of technical data that has been released and validated as adequate and complete for its intended purpose, as specified in MIL-STD-31000.

3.2.70 Validation status - reference technical data. A set of technical data that is provided for information, but is not necessarily validated as accurate, adequate, and complete, as specified in MIL-STD-31000.

3.2.71 Technology stack. A comprehensive environment capable of digitally modeling and massively iterating all its aspects without the shackles of real-world runtime, risk, or cost.

3.2.72 Threat modeling. A form of risk assessment that models aspects of the attack and defense sides of a logical entity, such as a piece of data, an application, a host, a system, or an environment. Additional guidance is specified in NIST SP 800-154 and NIST SP 800-53.

3.2.73 Virtualization. The act of creating a virtual version of something, including virtual computer hardware platforms, storage devices, and computer network resources.

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4. GENERAL GUIDANCE

4.1 MBE overview. The Model-based Enterprise (MBE) structure provides an illustration of various models that can be used to construct a Model-based Definition (MBD) of a system, sub-system, or component. The MBE may select multiple modeling types in an integrated and comprehensive approach to design, build, and sustain warfighter equipment. Digital Engineering (DE) provides an approach to engineering that uses models as an integral part of the technical baseline that includes the requirements, analysis, design, implementation, and verification of a capability, system, and product throughout the acquisition lifecycle. The program or project will ultimately select the models required to clearly communicate and define the capability or capabilities transitioned to the warfighter. The subtypes of modeling used by an MBE organization are referred to as MBxs. These MBxs are represented graphically on figure 1. Those presented are not intended to be all encompassing; other MBxs may be added to fit each organization's overall mission.

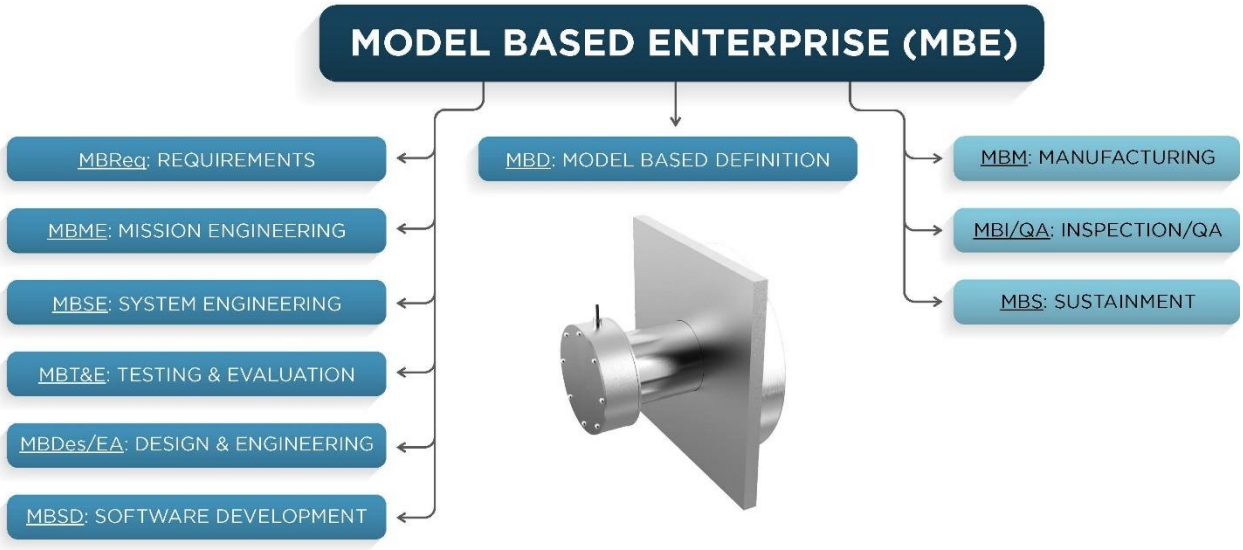


FIGURE 1. MBE structure.

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4.2 Benefits of MBE. An MBE provides the integration and collaboration environment to support the development and sustainment of systems and system of systems (SoS). Figure 2 is an example of an MBE that integrates MBD CAD models to perform MBSE and sustainment activities throughout the lifecycle of a system. Table I compares a model-based solution against the traditional approach.

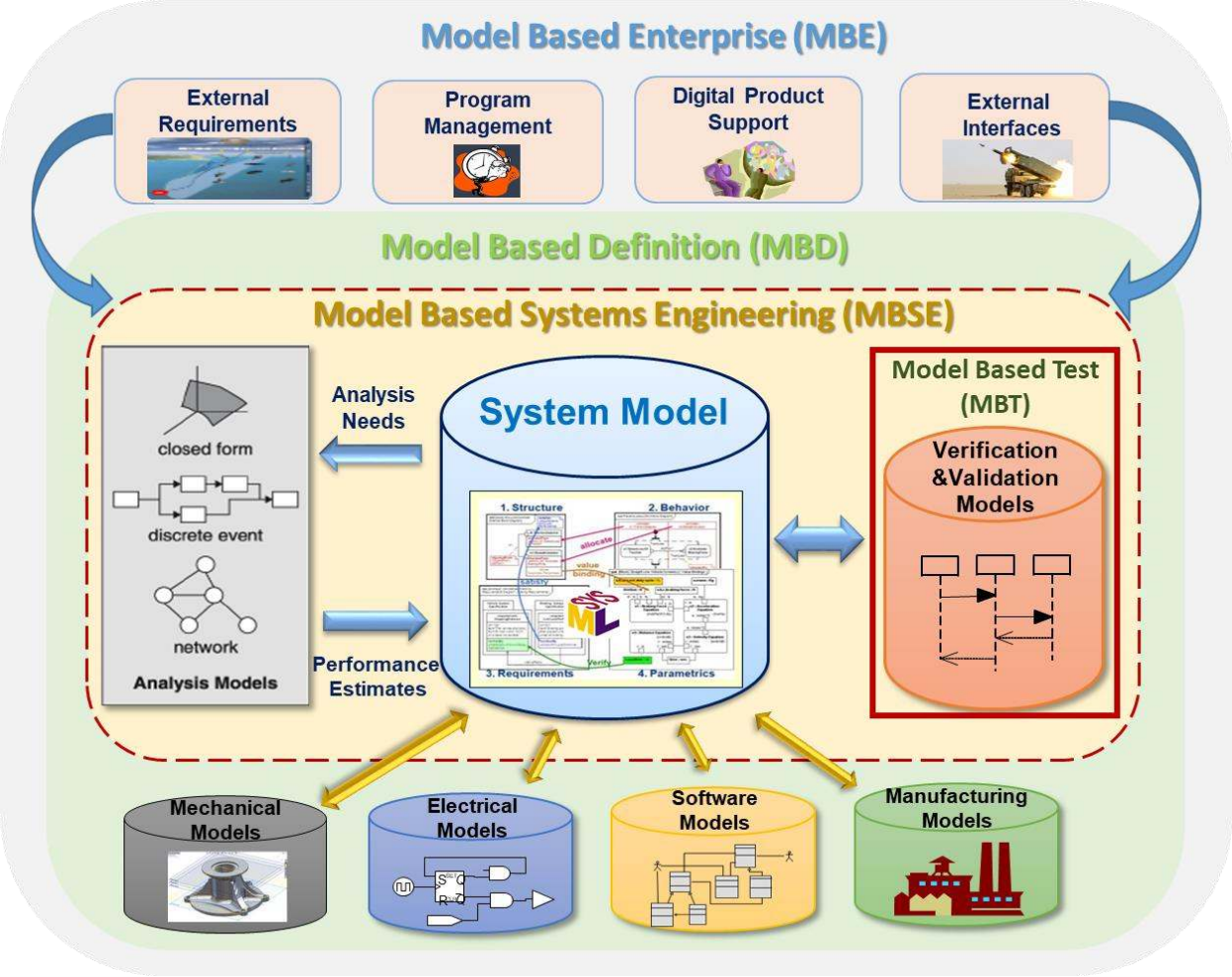
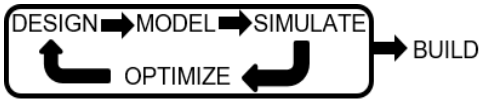


FIGURE 2. Structure of models to support MBE.

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TABLE I. Model based versus traditional enterprises.

Phase	Traditional Enterprise	Model-based Enterprise
Product Development	Design requirements established early in the design process make changes tedious and present challenges in keeping them updated.	Design requirements modeled and constantly reviewed and adjusted throughout the design process
	Design tradeoffs not well understood or considered	Design tradeoffs well understood, modeled, simulated, and constantly re-examined
	Design Process 	Design Process  and 
	Static design, development, production, and support processes	Constantly improving MBx capabilities and integration between MBx capabilities
Product Definition	Paper-based	Digital
	2D drawings represent the authoritative definition of the product	3D models authoritative definition of the product
	Focus on human-readable product data only	Focus on human- and machine-readable product data
	Engineering changes are slow and inefficient	Engineering changes are fast and efficient, and tailored to the context of the change
	Product definition changes done with hanging engineering changes that require the user of the data to have drawing and hanging changes in order to have the most up-to-date product definition	Product definition information is online, on-demand, and always up to date (i.e., changes are quickly incorporated into the product definition and no longer have hanging changes)
Project Information	Information travels as email attachments or hand carried hardcopies	Established digital workflows for standard processes
	Multiple copies of information and is difficult to version control	Information exists as a single authoritative master accessible by all authorized users and is easy to version control
	Information recreated and reentered for downstream applications	Information created once and accessed/linked by downstream applications
	Significant time spent looking for project information	Eight-second rule applies. Eight-second rule requires that all key product and project information (e.g., requirements, test plans, budget information, drawings/models) can be accessed within 8 seconds of the realization that it is needed

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TABLE I. Model based versus traditional enterprises – Continued.

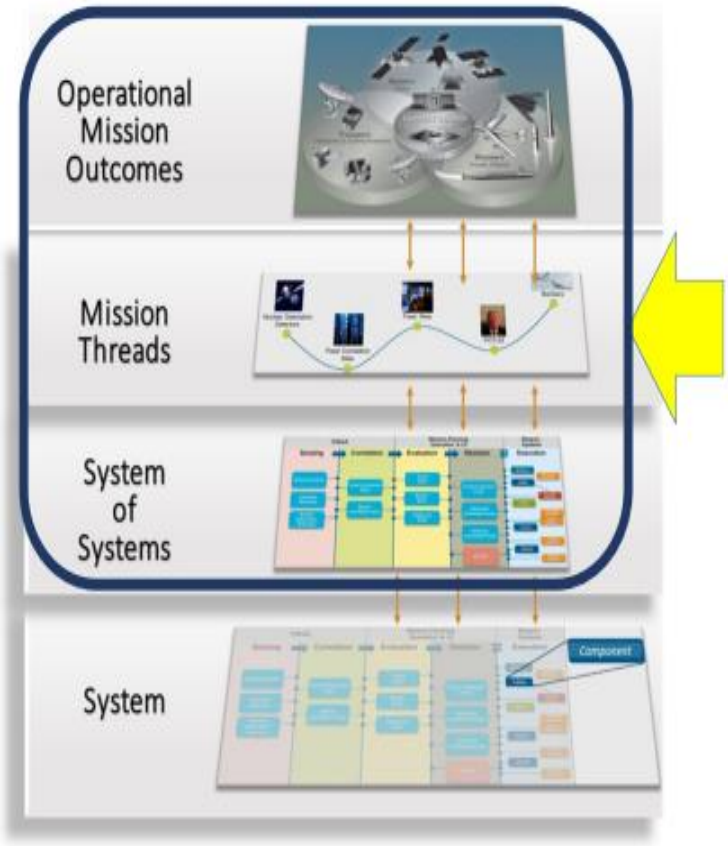
Phase	Traditional Enterprise	Model-based Enterprise
Project Information	Key project and product information cannot be relied on to be accurate or up to date. Local copies of key information are stored because official repositories are slow or unreliable	Key project and product information managed to ensure data visible, accessible, understandable, linked, trustworthy, interoperable, and secured (VAULTIS) in accordance with DoD data strategy (i.e., the data is accurate, people know it to be accurate and updated, and the information is traceable and controlled over its lifecycle).
Build Information	Design information separate from manufacturing information	Seamless flow from design to manufacturing
	Additive and advanced manufacturing an afterthought after design	Additive and advanced manufacturing built into the design process
	M-BOM created separately from E-BOM	M-BOM generated from and connected to E-BOM
Operations & Sustainment	Logistics and sustainment information separate from design	MBD information seamlessly flows into sustainment and logistics information
	P-BOM created and maintained independently	P-BOM generated from and connected to E-BOM
	Little feedback from the field on deficiencies or needed design improvements	Established and robust processes to obtain field feedback on product deficiencies and possible design improvements
	Product operational information (e.g., operator manuals and technical manuals/orders) consists of paper documents and black line art	Product operational information is digital, constantly up-to-date, and contains dynamic, interactive 3D models, how-to videos, and other multimedia presentations

4.3 DE SoS approach. For DE data to be available, discoverable, and useable, DE tools and repositories need to have electronic interfaces or the ability to effectively translate data between the program office platform and supporting activities. A program office will interface with mission engineering requirements as shown on figure 3. Operational mission outcomes use measures of effectiveness, and mission threads (MT) and mission engineering threads (MET) use measures of performance as metrics. This handbook is focused on supporting the program office, thus infrastructure requirements are not addressed in detail. These higher-level SoS details need to be considered along with the overall DE strategy and DE interfaces, as specified in the Department of Defense (DoD) DE Strategy and other department specific DE strategies, such as:

a. US Naval & Marine Corps Digital Systems Engineering Transformation Strategy
b. USSF Vision for a Digital Service
c. USAF Guidance for e-Program Designations
d. Air Force Digital Guide

DoD mission and SoS definitions can be found within the Mission Engineering Guide.

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Mission Engineering is the deliberate planning, analyzing, organizing, and integrating of current and emerging operational and system capabilities to achieve desired warfighting mission effects.

Mission Threads (MT) are the construct which links systems of systems (composed of systems and technology) to warfighting mission outcomes.

MT and Mission Engineering Threads link systems engineering to mission analysis

FIGURE 3. Authoritative mission-level architectures and MTs.

4.4 Model-based 3-dimensional (3D) product enterprise data. Moving from traditional 2-dimensional (2D) product data to 3D model-based data is a foundational part of DE. 3D model-based data is capable of providing complete data fidelity that can efficiently meet all product design requirements. At the same time, 3D data is more complex and requires more complex CAD tools, enhanced CAD hardware, and CAD training to effectively manage the data. Establishing this infrastructure is a required step in the process. Once the infrastructure is in place, the 3D model-based data can serve as a master source and drive process efficiencies with the elimination of redundant repositories, duplicate data, file translation issues, and unusable or lack of data for government activities. When properly managed within an infrastructure as a service (IaaS) or a platform as a service (PaaS), the data will be made available when and where it is needed, discoverable with what is needed, and useable for all use cases.

4.4.1 Model-based data. Model-based data in the form of tables may be more efficiently managed outside of the CAD tool via a product lifecycle management (PLM) interface. Many different data types and formats can exist within the PLM interface and improved accessibility and data sharing is available. Data formats are shown on figure 4.

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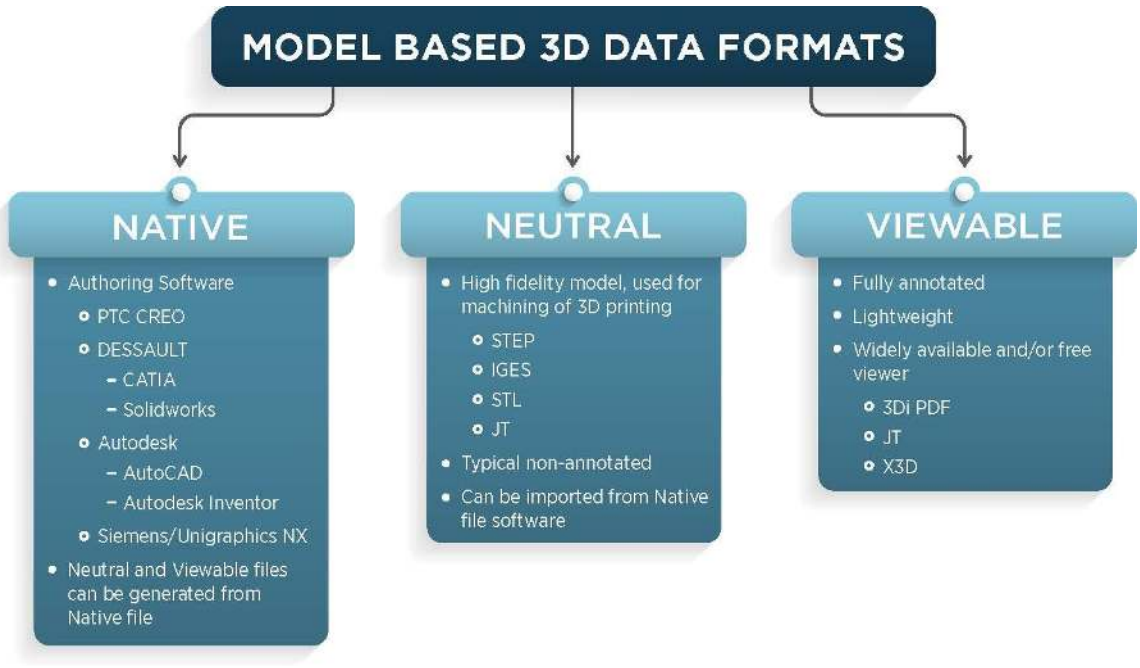


FIGURE 4. Model-based 3D data formats.

4.4.2 3D model authoritative source. The 3D model is developed using a CAD tool and becomes the master within a CAD tool. When this data is shared outside of this master source, it becomes a different type of data that will be connected to the master via a digital thread. Data types that are connected with the digital thread are shown on figure 5.

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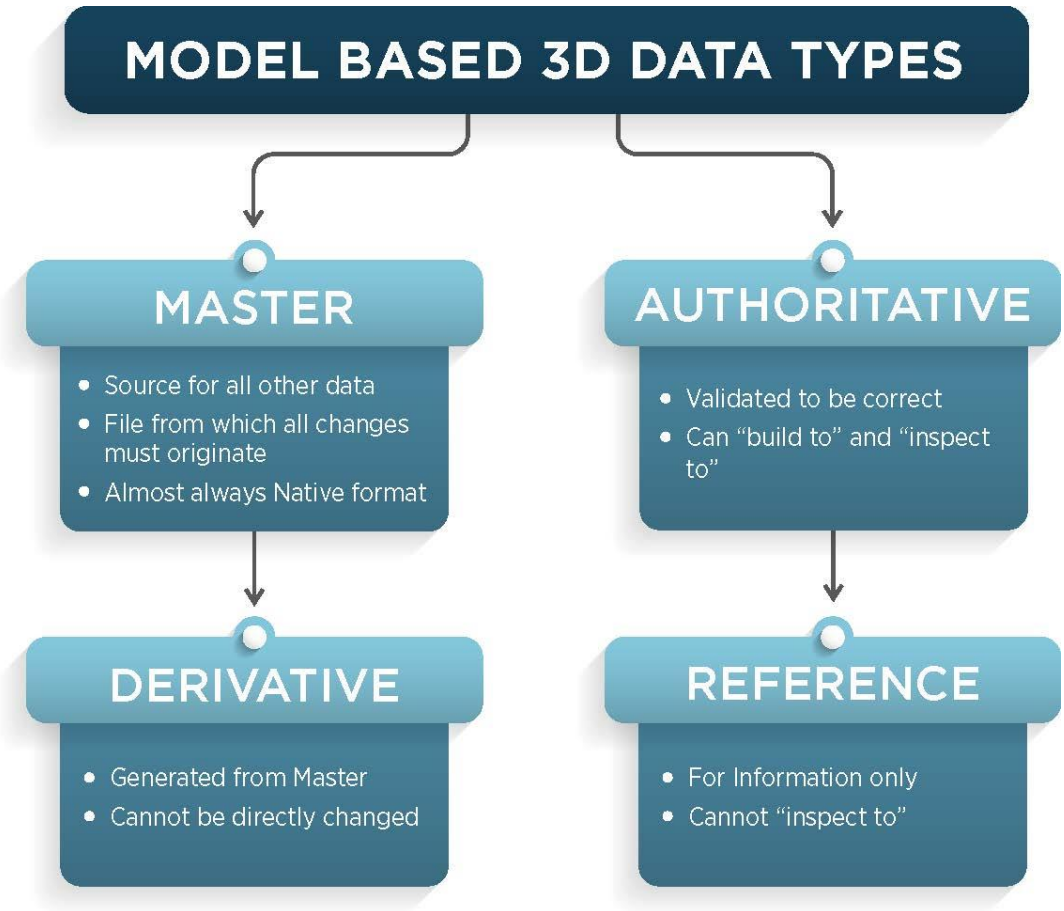


FIGURE 5. Model-based 3D data types.

4.4.3 PLM. The CAD modeling tools can effectively interface with these other data forms (e.g., tabular data) using PLM tool software. By using the PLM interface, the various forms of data can be managed using the best software tool. PLM tools are better equipped to handle data types such as tables, bill of materials (BOM), or reports. Each program office needs to ensure that any digital interfaces established and used allow for data interchange through the acquisition lifecycle.

4.4.4 CAD data exchange. Data flows along the digital thread. When data is moved from one system to an interfacing system, it can lose a portion of its complete data fidelity. The only way to transfer a complete data package is by transferring native data. When different CAD tools or environments are used, the data may need to be transferred using neutral or some other type of viewable data format. If a preferred solid model data native format is unavailable, additional guidance is specified in ISO 10303 and NSRP ASE 2009-350. It is essential that each interfacing activity define the minimum data requirements to ensure that the necessary data is effectively transferred to meet their requirements. Additional industry standards for data exchange can be found in Appendix H.

4.5 Lifecycle data requirements. Using an MBSE approach, data and information can be mapped so that data flows throughout the organization and throughout PLM. Various stakeholders will need access to various views or elements of data. For this process to work effectively, user requirements need to be known and planned for up front in the acquisition process. MBSE planning includes and supports the various lifecycle activities as shown on figure 6. On this figure, the digital thread is represented by the blue arrows that flow through all parts of the design, build, and sustainment process of a typical aircraft carrier lifecycle. This approach utilizes authoritative product models to design a product or system and makes use of the design information for all downstream processes such as build, manufacture, and logistics. To meet all user requirements, the data needs to be continually retrievable and mapped for access by required users in this process.

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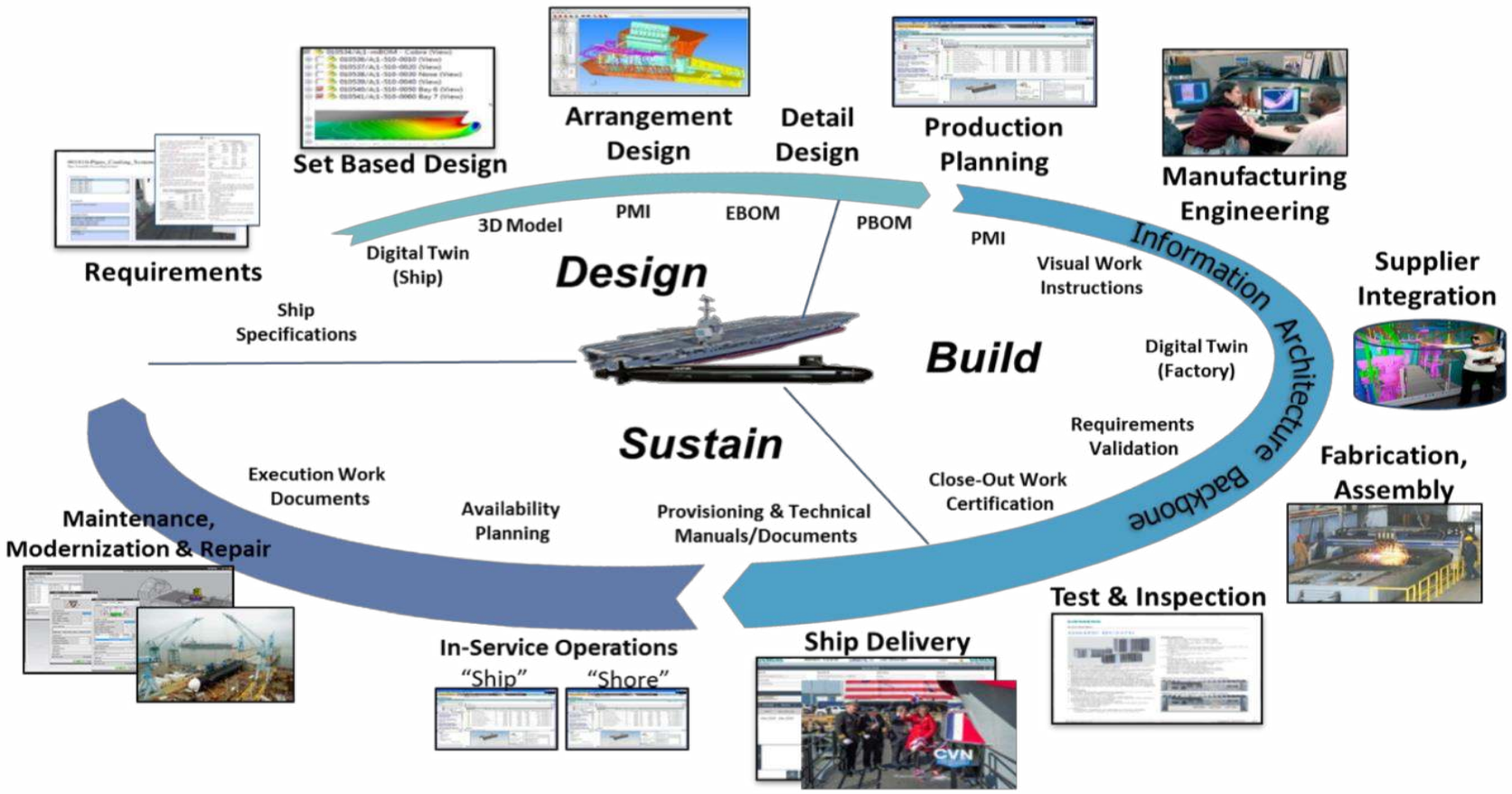


FIGURE 6. DE data maturation lifecycle example.

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5. DETAILED GUIDANCE

5.1 Data management evolution. A summary of six basic stages of incremental DE data management ecosystems is shown on figure 7. These six stages show a progression of moving from paper documents to a full digital environment. Stage 1 is a prime contractor-controlled technical data package (TDP) with 2D paper driven outputs as shown on figure 8, and the final stage (stage 6) is a Government-controlled TDP with 3D model-based data ownership, as shown on figure 13 (and as specified in DoDI 5000.93). As a program office progresses through these stages, it can achieve process efficiencies and cost savings with data management. For each subsequent stage in the DE progression, the Government establishes the data type and data format requirements to acquire more authority to view, access, and modify the master data. Each program office should review which stage of digital data provides the best value for their program. Solutions will need to be tailored based on infrastructure and stakeholders, as well as cost and schedule constraints. The data includes the TDP, as specified in MIL-STD-31000, data analytics and simulation, and the development of acquisition data products (e.g., data item descriptions [DID], CDRLs), as specified in MIL-STD-963.

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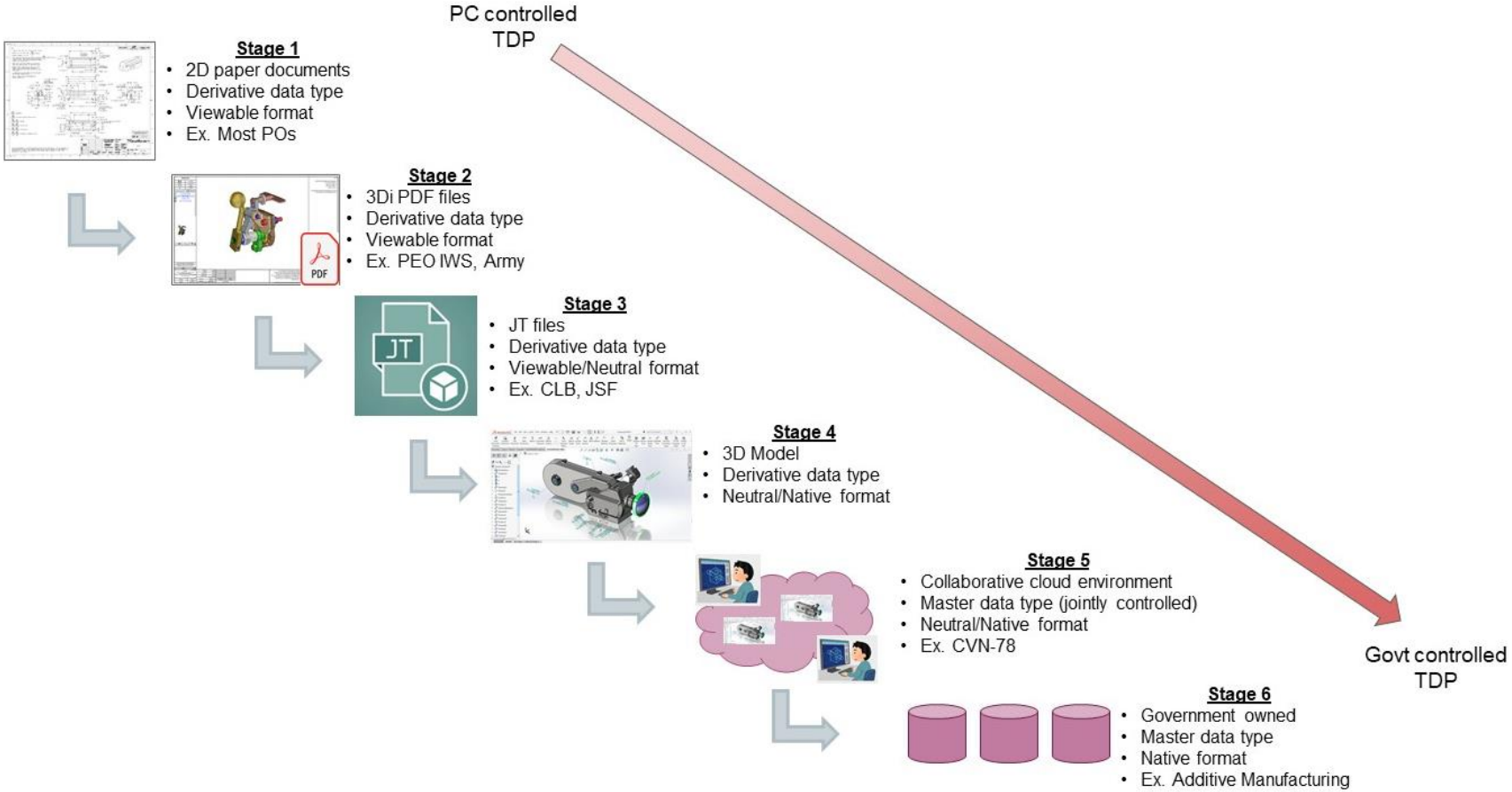


FIGURE 7. Example of DE data evolution stages.

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5.1.1 Incorporating DE capabilities. Each program office should periodically assess and identify their current data management approach. This handbook can be used as guidance to transition to the DE stage, or ecosystem, and to support cost, schedule, and performance requirements. Incorporation of advanced DE stages have resulted in quantifiable savings to program teams. Cost savings include more efficient maintenance guidance, development process sheets or production illustrations, and maintenance cards.

5.1.2 DE data evolution stages. The various stages will have product drawing classifications as specified in ASME Y14.100 Appendix F, Classification Codes for Drawings and Data Sets. Stage 1 refers to a classification code 2 data set with a model and drawing graphics sheet. The 2D drawing will contain master data elements. Stages 2 through 4 refer to a classification code 3 data set with a model and simplified drawing graphics sheet. The 3D model becomes the master source for geometric part attributes. Stages 5 through 6 refer to a classification code 5 data set with a model and no drawing graphic sheet. The 3D model becomes the master source of product data. The applicability of the various data attributes by progressive evolution stages is shown in table II.

TABLE II. DE data progression by evolution stage.

Data Attribute	Stage 1	Stage 2	Stage 3	Stage 4	Stage 5	Stage 6
2D Drawings Delivered to Govt.	X	X	X	X	---	---
3D Viewable Drawings Delivered to Govt.	---	X	X	X	X	---
3D Neutral Models Delivered to Govt.	---	---	X	X	X	X
3D Native Models Delivered to Govt.	---	---	---	X	X	X
Master 3D Model Data Managed by the Contractor	X	X	X	X	Partial	---
Master 3D Model Data Jointly Controlled by the Contractor and Govt. Collaboratively	---	---	---	---	X	---
Master 3D Model Data Managed by the Govt.	---	---	---	---	Partial	X
Master 3D Model Data Developed and Managed by the Govt.	---	---	---	---	---	X
ASME Y14.100, Appendix F, Classification Codes and Data Sets	Code 2	Code 3	Code 3	Code 3	Code 5	Code 5

5.2 Stage 1 – data access of traditional 2D documents. Traditional product data approaches have used stage 1. Data delivery of legacy 2D drawings, as seen on figure 8, is typically contractually invoked between the prime contractor and Government program office. The prime contractor typically has ownership of and manages the master data, while the Government has access as detailed in the data rights section of the agreed contract. The program office should determine the program data rights in the best interest of the Government in order to maintain configuration and data management.

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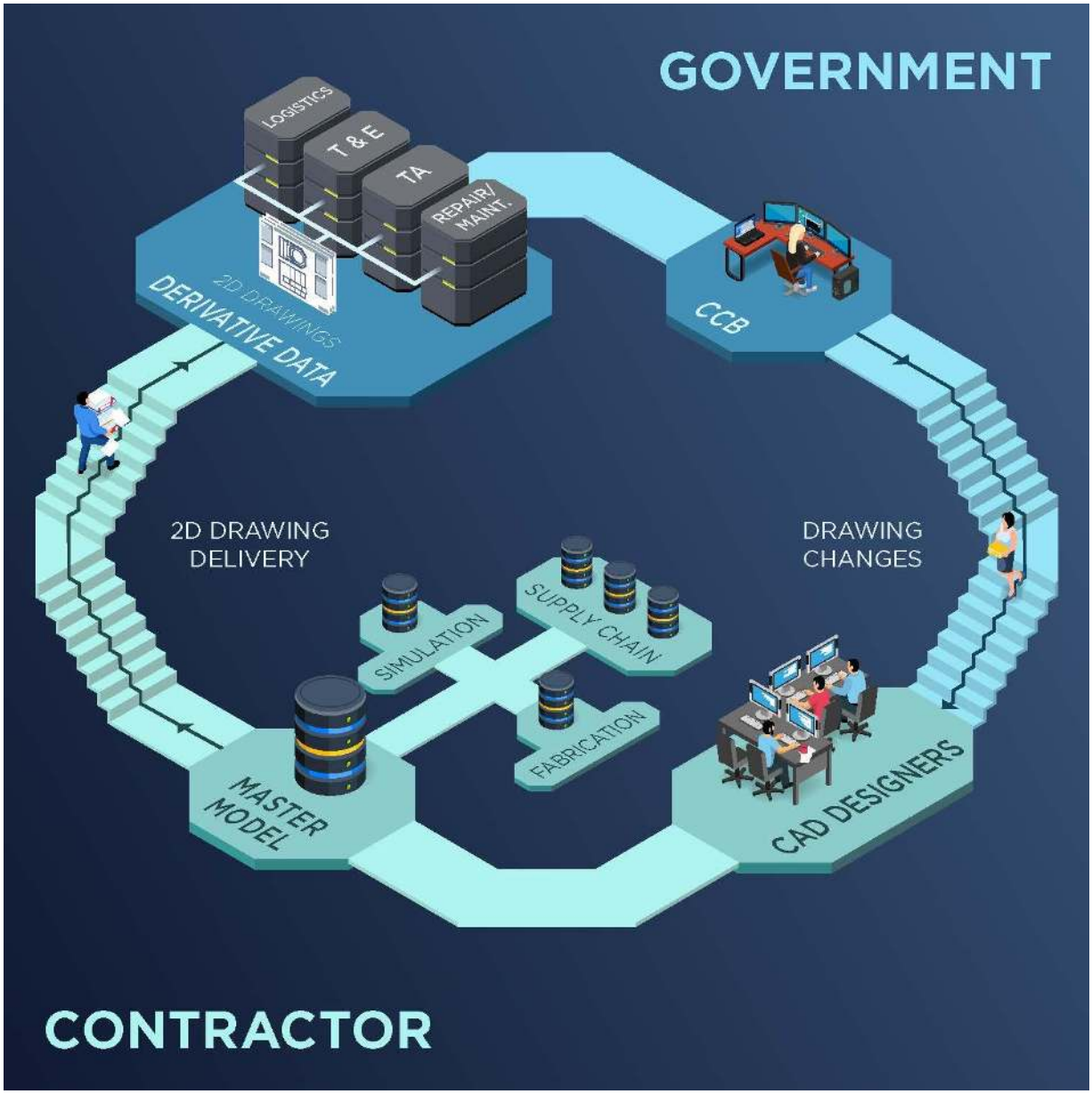


FIGURE 8. Stage 1 – data access of traditional 2D documents.

5.2.1 Stage 1 master model. The master model is typically controlled by the prime contractor in their digital environment. To create 3D models configurationally controlled by the prime contractor, derivative datasets pulled from the authoritative product model are made available to the Government in the form of 2D paper drawings. Traditional 2D paper documents, such as 2D assemblies or shells or 2D product illustrations, are transferred to various Government repositories owned by various program offices and Government end users to support logistics and in-service entities. This stage is common in most Government program offices. Requested changes from either side can be built into the existing 3D master model after approval of the configuration control board (CCB). Any proposed changes from the Government side are sent to the Government CCB for approval, and then to the prime contractor CCB. If the change is approved, a prime contractor CAD designer will make the change in the master model. Proposed changes from the prime contractor are either sent to the Government CCB for approval, or directed straight to the prime contractor CAD designer, depending on where the technical authority to approve the change lies.

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5.3 Stages 2 – data access with 3-dimensional intelligent (3Di) data delivery. Stage 2 includes the delivery of 3Di models that are representations of the native 3D models that can be viewed on a personal computer. The 3Di models provide a greater capability to the user over the traditional 2D drawings, as shown in figure 9. The file format used can vary from program office to program office. Derivative viewable 3Di portable document formats (PDF) are generated from a prime contractor managed 3D model. For example, sections of Program Executive Office Integrated Warfare Systems and the Army have transitioned to using 3Di PDFs. Other formats such as the hypertext markup language (HTML) can also be used.

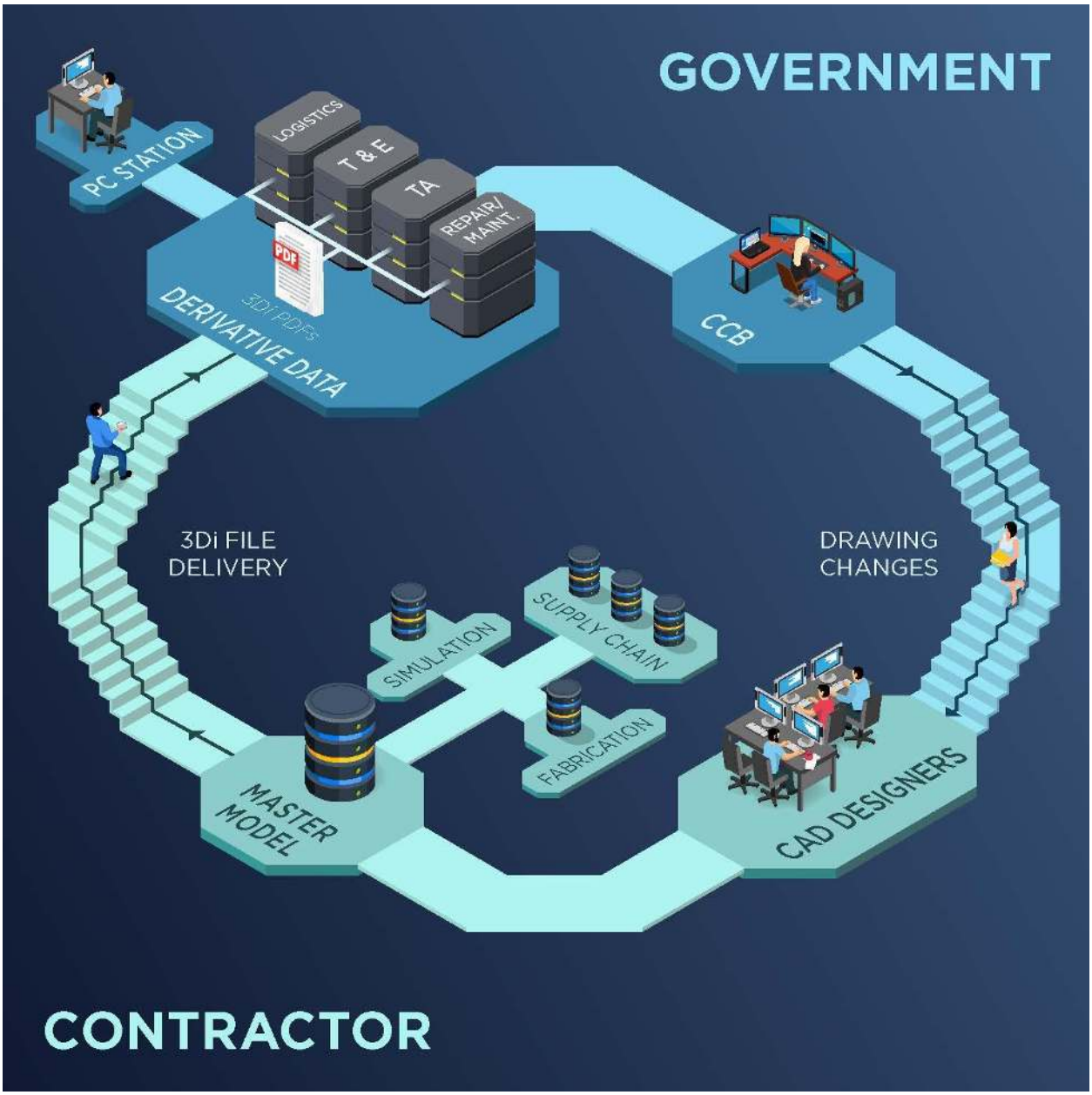


FIGURE 9. Stage 2 – data access with 3Di data delivery.

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5.4 Stage 3 – data delivery of Jupiter Tessellation (JT) files and 3D models. Stage 3 includes the data delivery of JT files and the 3D models as shown on figure 10. The 3D model formats are contractually invoked with the prime contractor. Depending on the structure of the contract, the prime contractor may still own and manage the master data. The overall drawing requirements for government products will be determined by the corresponding drawing deliverable approval procedure . The DID will define the data format of deliverables from the prime contractor to the Government end users, as specified in DoDM 4120.24. Derivative JT files provide data in a viewable and neutral format. Similar to a combination of a 3Di model and a neutral STEP file, JT files are delivered as a 3D model with other details included in a PDF. The JT file is viewable, lightweight, and can also be used in downstream manufacturing.

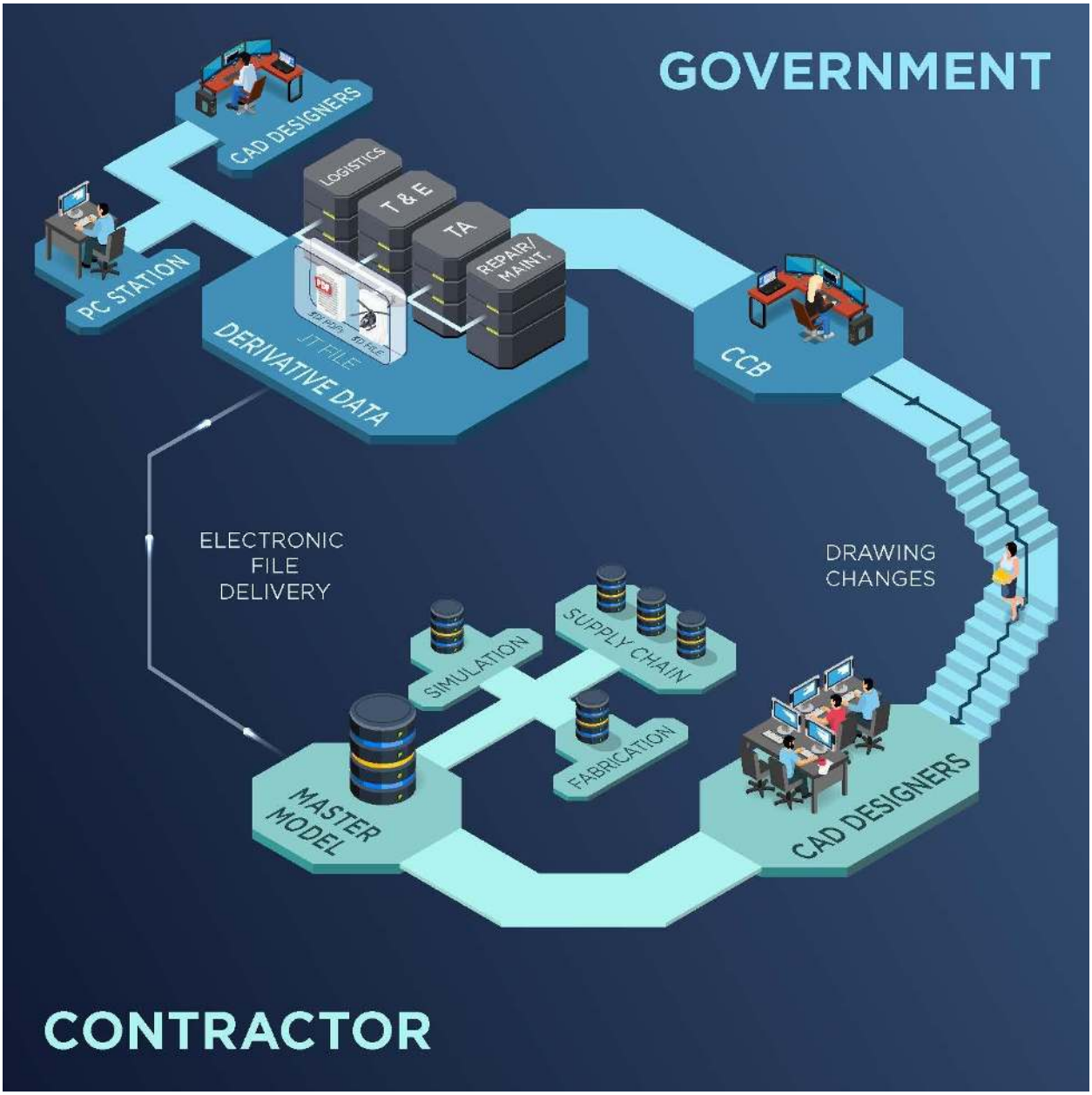


FIGURE 10. Stage 3 – data delivery of JT files and 3D models

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5.5 Stage 4 – data delivery of native 3D models. Stage 4 includes the data delivery of the 3D models as shown on figure 11. The derivative 3D models can be in a neutral or native format. In order for the government to manage a master data set, the 3D models should be in a native format.

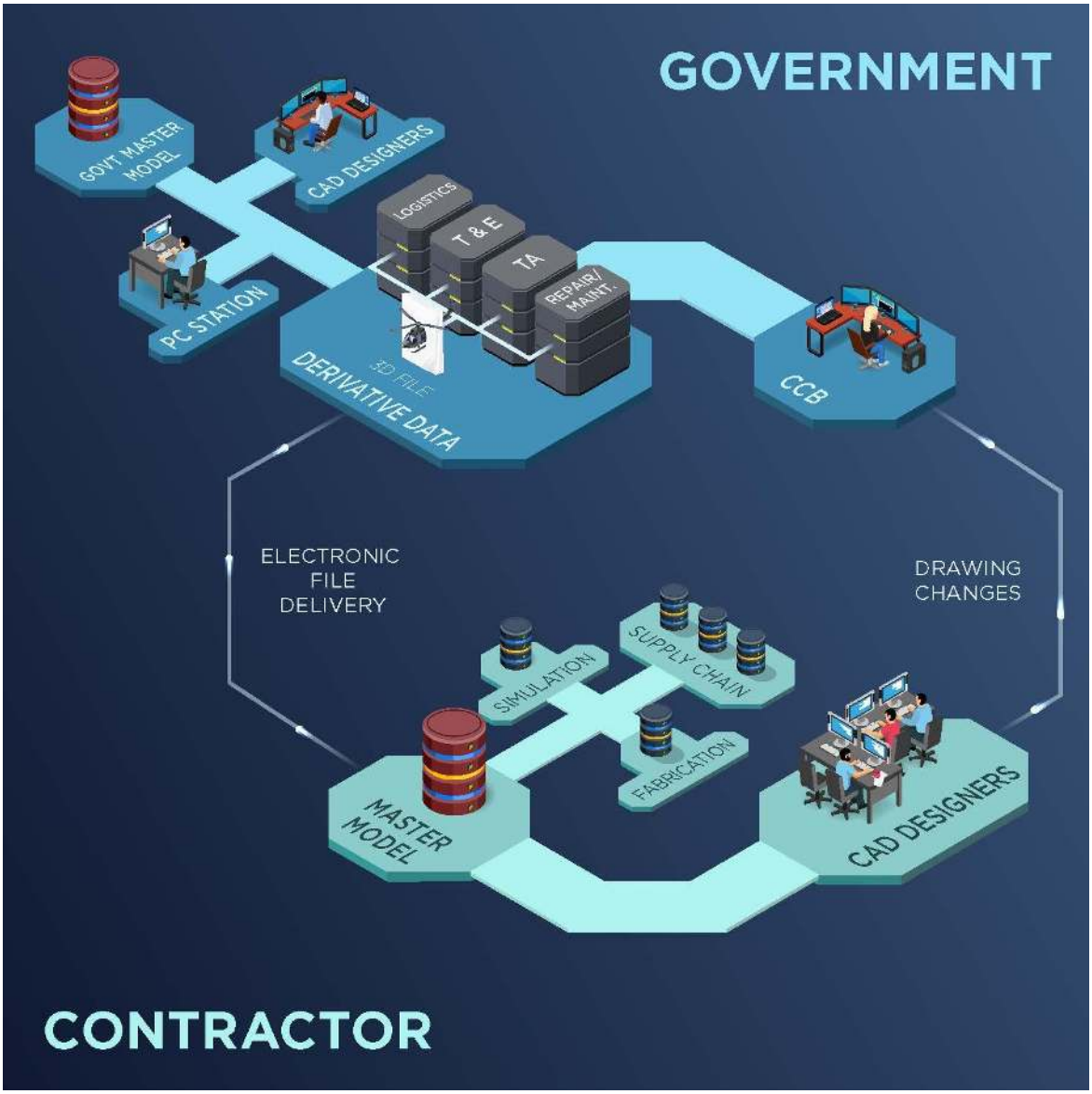


FIGURE 11. Stage 4 – data delivery of native 3D models.

5.5.1 Master model data sharing. The 3Di PDF, JT, and 3D approach of sharing data from the master model saves costs in comparison to the traditional 2D approach. The production of these model files incurs more upfront costs, but once it is produced it can be used for every function that requires derivative data produced from the 3D model. Use of exclusively 2D data is more expensive because each source will require their own copy produced specifically to their standards. Any changes are then relayed and repeated through several different documents. Drawings are also more frequently lost in the 2D format. By the time each of the sources obtains a copy of the 2D data using the traditional method and all changes have been made, the total time and money spent far exceeds the cost of using the 3Di PDF, JT, or 3D model. Requested changes are still ultimately made by a prime contractor CAD designer.

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5.6 Stage 5 – collaborative cloud environment. In stage 5, the master data and 3D models previously owned by the prime contractor are stored and modified in a jointly-controlled cloud environment, as shown on figure 12. Ownership of the cloud environment could vary, but most likely belongs to a third party. Both the government and the prime contractor have their own CAD designers. The government CAD designer or Technical Warrant Holder (TWH) can recommend and redline changes while the prime contractor integration CAD designer can make changes. As an example, Ford class aircraft carriers use this management approach. Changes recommended by government entities are reviewed by the applicable TWH, redlined, and put into the collaborative cloud environment, but are not yet incorporated into the master model. The changes are viewed in the cloud environment by the prime contractor CCB. If approved, the prime contractor integration CAD designer builds the changes into the master model.

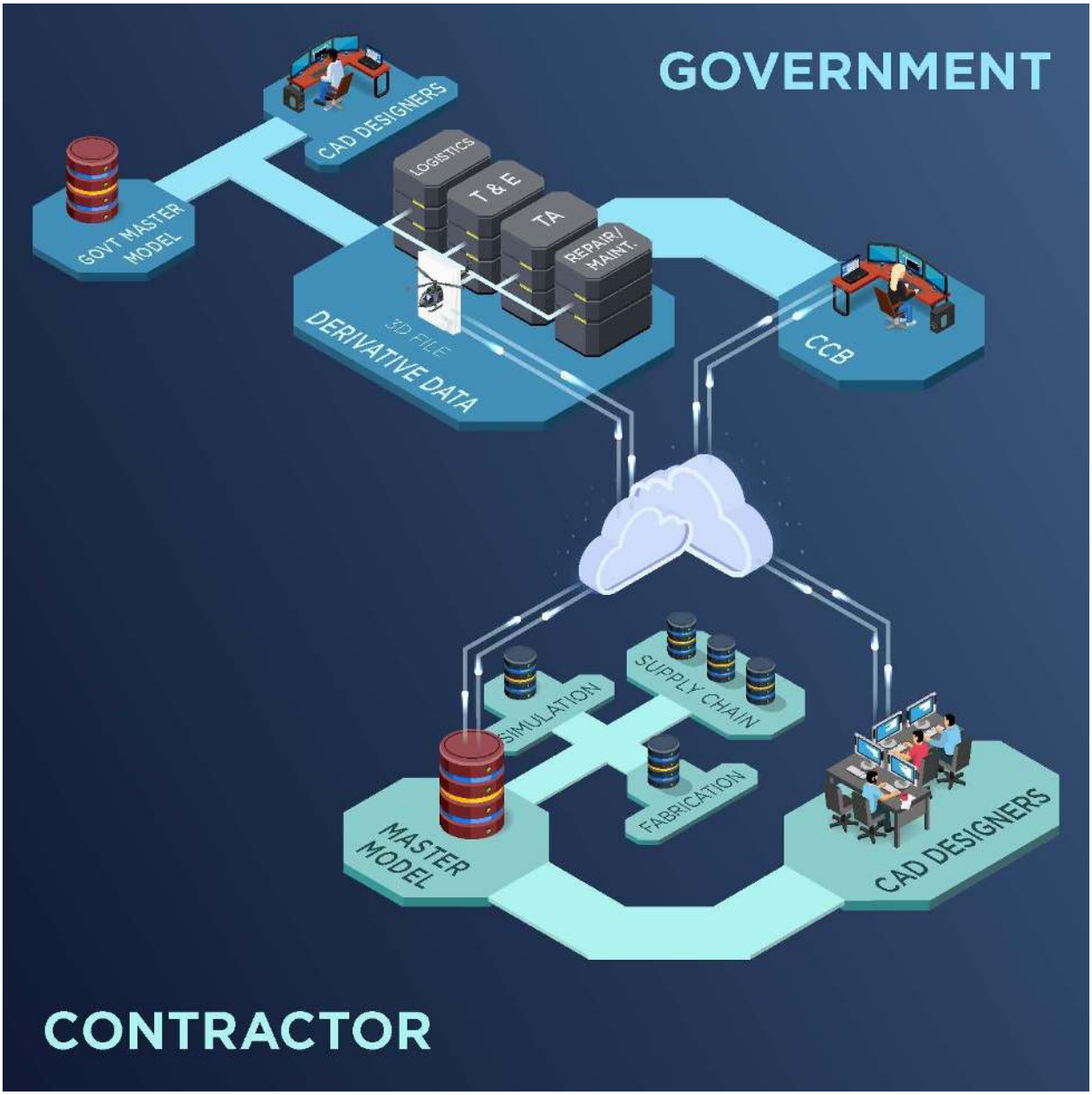


FIGURE 12. Stage 5 – collaborative cloud environment.

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5.7 Stage 6 – Government-owned master data. In stage 6, the Government would ideally have complete control of program integrated logistics support activities, as seen on figure 13, and the Government owns the master data set. Government CAD designers would manage all master product changes and directly support all government requirements using a government repository. This stage offers the greatest benefit to DoD with PLM and sustainment requirements. The increased control and access to product data reduces PLM contracting costs and allows the DoD to respond quickly with evolving threats.

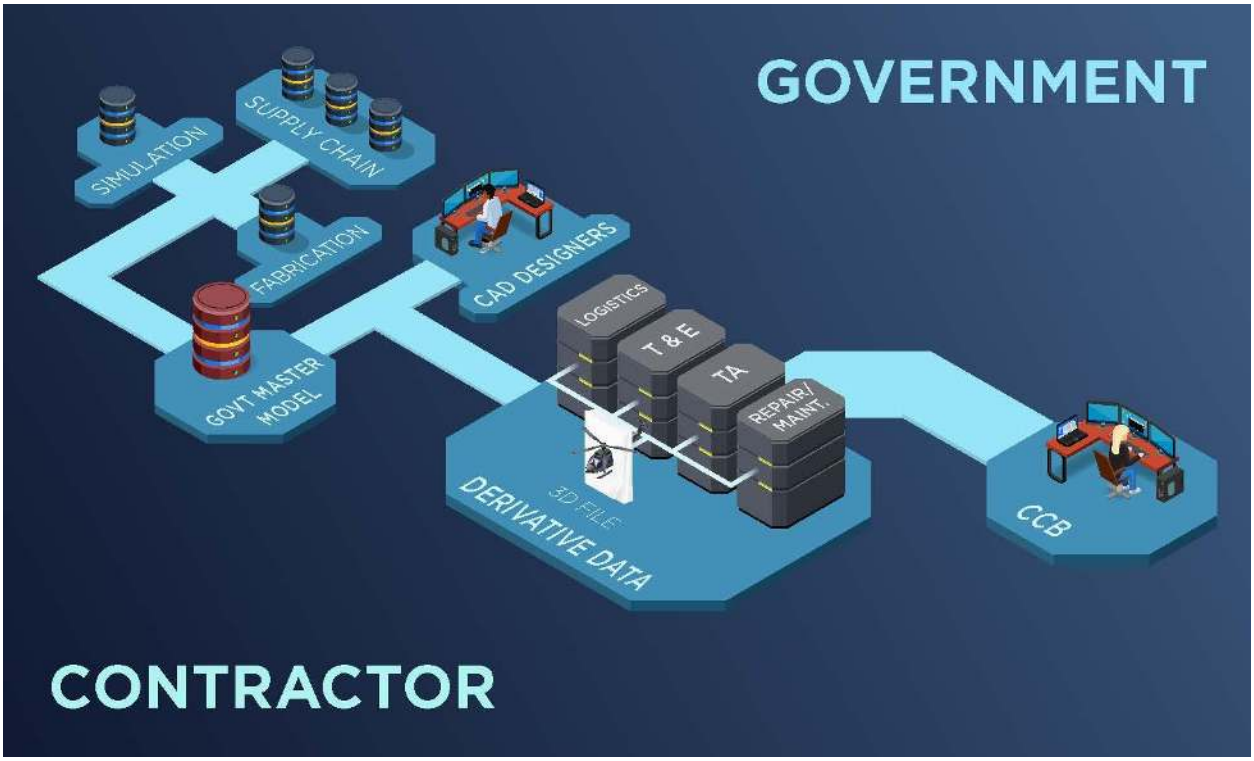


FIGURE 13. Stage 6 – government -owned master data.

5.8 Intellectual property (IP) considerations. When determining what standard or specially negotiated license the Government needs to acquire the delivered models, the Government will accurately identify the following:

a. All entities that need to receive copies of the model and by what means (e.g., physical, electronic)

b. How those entities will use those models

c. How long those entities will use those models

These are the parameters that establish the scope of any IP license to satisfy the program's intelligence threat inputs/supportability considerations, as specified in DoDI 5000.86, and product support strategy, as specified in DoDI 5000.91.

6. ACQUISITION GUIDANCE

6.1 General. Effective DoD acquisition requires proper planning and strong partnerships with contractors. DE supports all adaptive acquisition framework pathways as specified in DoDI 5000.02. Additional DoD guidance is specified in DoDI 5000.75 and DoDI 5000.88 for topics covered in this handbook.

6.2 Acquisition planning. An effective government MBE requires up front planning. Two important elements are the DE and MOSA statutory requirements, which are closely interrelated. Both are included within the FY21

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National Defense Authorization Act as DoD congressional reporting requirements. The five MOSA pillars, as shown on figure 14, should be included in acquisition planning. Programs should address their DE and MOSA plans in their acquisition strategies and as specified in the systems engineering plans (SEP).



FIGURE 14. Five MOSA pillars.

6.3 Five MOSA pillars. The five pillars include the following elements:

a. Establish enabling environment or integrated digital environment (IDE) (see figure 15)
   (1) Integration of development and operations
   (2) Phased technology insertion by module
   (3) Cloud data sharing
b. Employ modular design
   (1) Cohesive, encapsulated, self-contained, highly binned
c. Designate key interfaces
   (1) Published key interfaces
d. Select open standards
   (1) Well defined, mature, widely used, readily available
e. Certify conformance
   (1) Published conformance criteria
   (2) Automatic testing and certification

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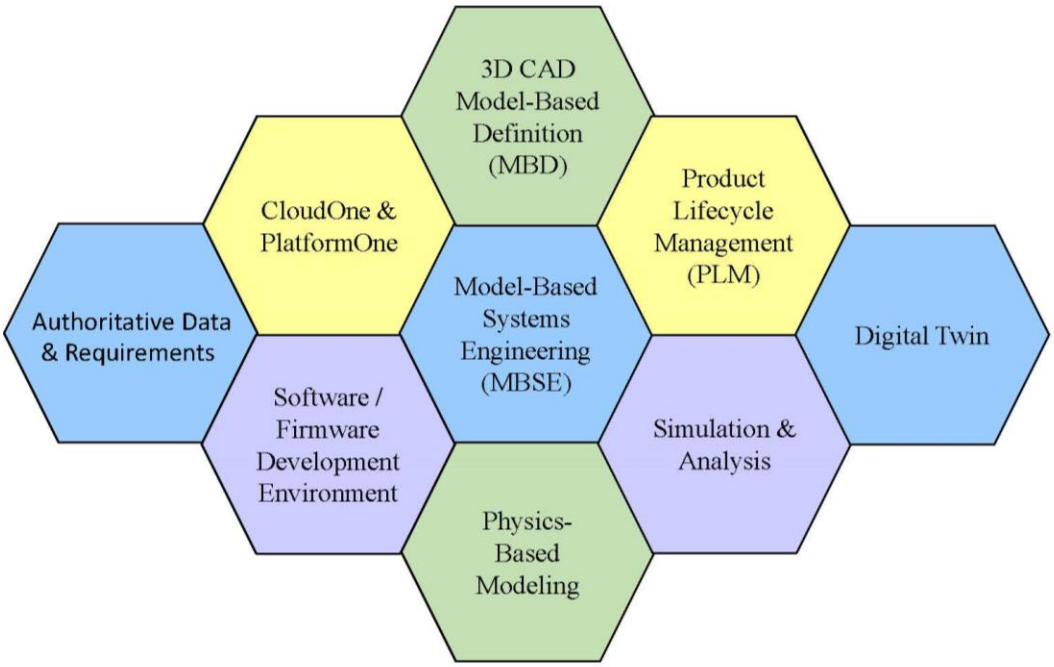


FIGURE 15. Elements found in an IDE.

6.3.1 DE MOSA pillars. Implement DE early in the first three systems engineering technical processes (stakeholder requirements definition, requirements analysis, and architecture design), where mandated defense information technology (IT) standard registry (DISR) standards, risk management framework (RMF) cybersecurity controls, and system level continuous monitoring (SCLM) strategies are incorporated into the functional baseline architecture. The published functional baseline architecture includes RMF step one "categorization" (confidentiality, integrity, and availability), services view (SvcV)-6 (system data exchanges, measures, and metrics), standards view (StdV)-1 , and StdV-2. StdV-1 are mandated DISR standards and StdV-2 are emerging standards applicable to the functional baseline architecture. Interested programs planning to reuse published functional baseline architectures should update the SvcV-6, StdV-1, and StdV-2 Department of Defense Architecture Framework (DoDAF) views to reflect the current DISR approved baseline. MOSA-compliant DE supports the interoperability and supportability of new capabilities for existing programs and new start programs. The development of the functional baseline architecture can then be integrated into the existing program's digital system model (DSM) or form the foundation for a new start program. MOSA-compliant architectures can be reused by another interested program seeking to field the same capability or similar feature set. MOSA ensures IT standards compliance and assures capabilities old and new remain interoperable and supportable.

6.4 Model variants. Different model variants or use cases are available to help with the digital transformation of manual, document-centric processes being performed within the Government. The most important model is the DSM that represents the physical system of focus, also known as a digital twin, and the authoritative TDP and related data management artifacts. Two widely used variants of models are CAD models and MBSE models. CAD models support 3D product and manufacturing information. MBSE models support system engineering efforts across the acquisition lifecycle of a system. There are various modeling tools that can be used to create the TDP, and they have an associated suite of other input and output model-based tools that can also be used within the larger MBE environment.

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6.5 Modeling tools. There are several digital modeling tools available today. These tools provide a collaborative platform for research, design, development, and verification methods that improve cost, schedule, and performance of the acquisition item. Every effort should be made to conduct an exhaustive review of vendors and the digital modeling tools they offer, which will support selecting the tool that best meets the programs need. These tools also form the basis to support more efficient sustainment and modification of acquisition items that are addressed in other chapters of this handbook. It is recommended to use an order of preference with tool selection. Some tools can share data better than others. A preferred list of tools will minimize the proliferation of tools and provide simple, more efficient translation of data. Tools based on different standards or design criteria will make translated data files larger and result in a greater loss of data fidelity. A sample order of preference is shown in Appendix B. Before establishing an order of preference, the Government will verify that it has acquired the appropriate seat and site license to those tools under another contract so that the Government will be able to view, using its IT infrastructure, the models the contractor will develop and deliver via the TDP CDRL.

6.6 Physical models. The TDP is driven by the physical model. Physical models will form the focal point with other modelling inputs (e.g., MBReq) and outputs (e.g., MBT&E) within MBSE. The physical model will involve both hardware and software elements. Physical modeling will help create both the hardware and software modules, and SoS integration models.

6.7 DoD physical modeling and standards. Standards can also be included within digital environments, and not just as 2D paper documents. For example, the DoD is developing a standardized approach for modeling unmanned systems (UxS) using the information exchange standards shown on figure 16 to support DE and MOSA.

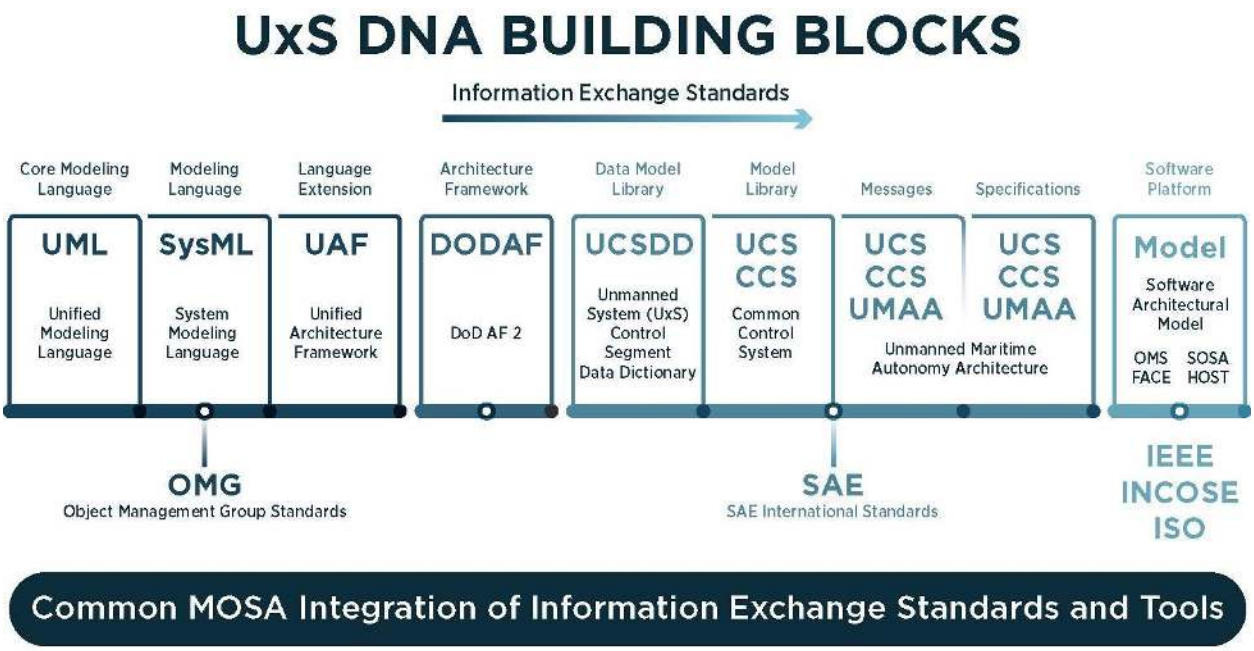


FIGURE 16. UxS modeling building blocks.

6.7.1 Translation cost drivers. The initial modeling languages shown on figure 16 are specified in the OMG unified modeling language (UML®) and OMG systems modeling language (SysML ™) standards. Standardization of a modeling language will achieve improved interoperability. When a model is delivered in other formats, there will be a cost to convert the model into another modeling tool. Translation can be costly and detailed analysis is recommended in selecting the best tool or tools available.

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6.7.2 Model metadata interchange. The Object Management Group (OMG) standard for exchanging metadata information via extensible markup language (XML) is the XML metadata interchange (XMI). XMI enables interchange of metadata between meta object facility based metadata and UML®-based modeling tools. Vendor XMI implementations have technical challenges exchanging UML® model files between tools. Programs should standardize modeling tool sets early in development and carefully consider costs to move data and files between tools. Tool data interchange incompatibilities should be analyzed when new tools are desired for use. New tool analysis should be performed in-terms of program advantages versus associated costs and effects to the schedule.

6.7.3 Data dictionary. One of the standards being used across multiple programs is the data dictionary, which can be applied to all DoD systems. The data dictionary is a glossary of terms that were designed to support all DoD acquisitions. The use of digital standards provides efficiency in the acquisition process and improved integration between multiple activities. As an example, a low-level data dictionary used across DoD is specified in SAE AS-6969.

6.8 DoDAF models. DoDAF is an evolving model description that allows DoD to share information more effectively. DoDAF focuses on architectural data and does not define a physical model. This data can consist of documents, spreadsheets, dashboards, or other graphical representation templates and is defined as a “view”. A collection of views is defined as a “viewpoint”. The viewpoint is the DoDAF described model.

6.9 Standardization in acquisition to support logistics. Sustainment of an acquisition item is the largest cost in the acquisition lifecycle and needs to be addressed within the development process. Proper planning to minimize the number of tools being used, a common work breakdown structure, and standardized modeling approaches will reduce the logistical footprint. DoD has standardized the DE logistics structure within the services, as shown on figure 17. When differences exist, the data translation and exchange procedures can be addressed to minimize logistic costs. Additional guidance is provided in Section 7.

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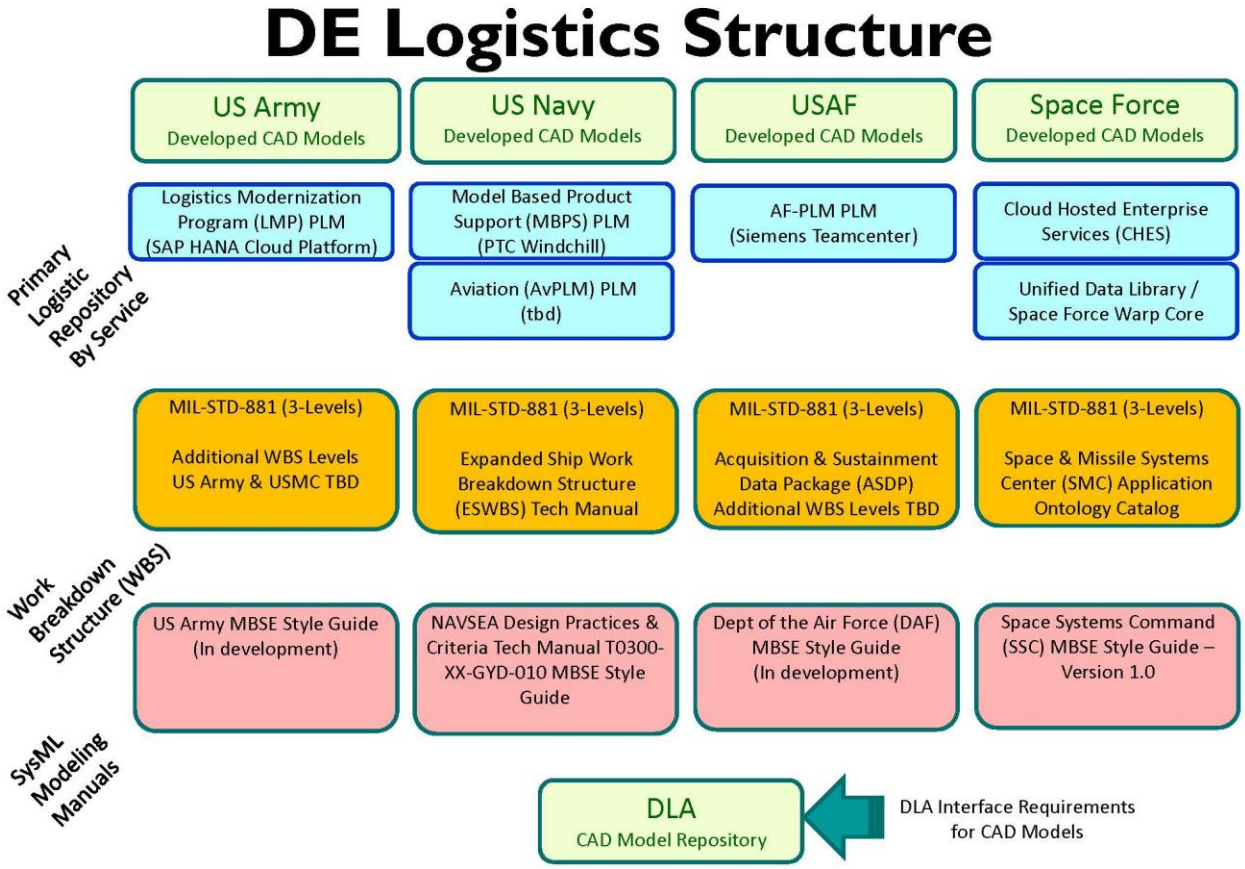


FIGURE 17. DE logistics structure.

7. PLM AND PRODUCT SUPPORT GUIDANCE

7.1 General. PLM refers to the management of data and processes used in the design, engineering, development, production, deployment, operation, support, and disposal of a product (weapon system) across its entire lifecycle and across the supply chain. The term PLM has historically been associated with manufacturing, but is currently most often associated with a software solution that integrates stakeholders with supporting data, process, and business systems in a computerized enterprise environment. The PLM system overview is shown on figure 18.

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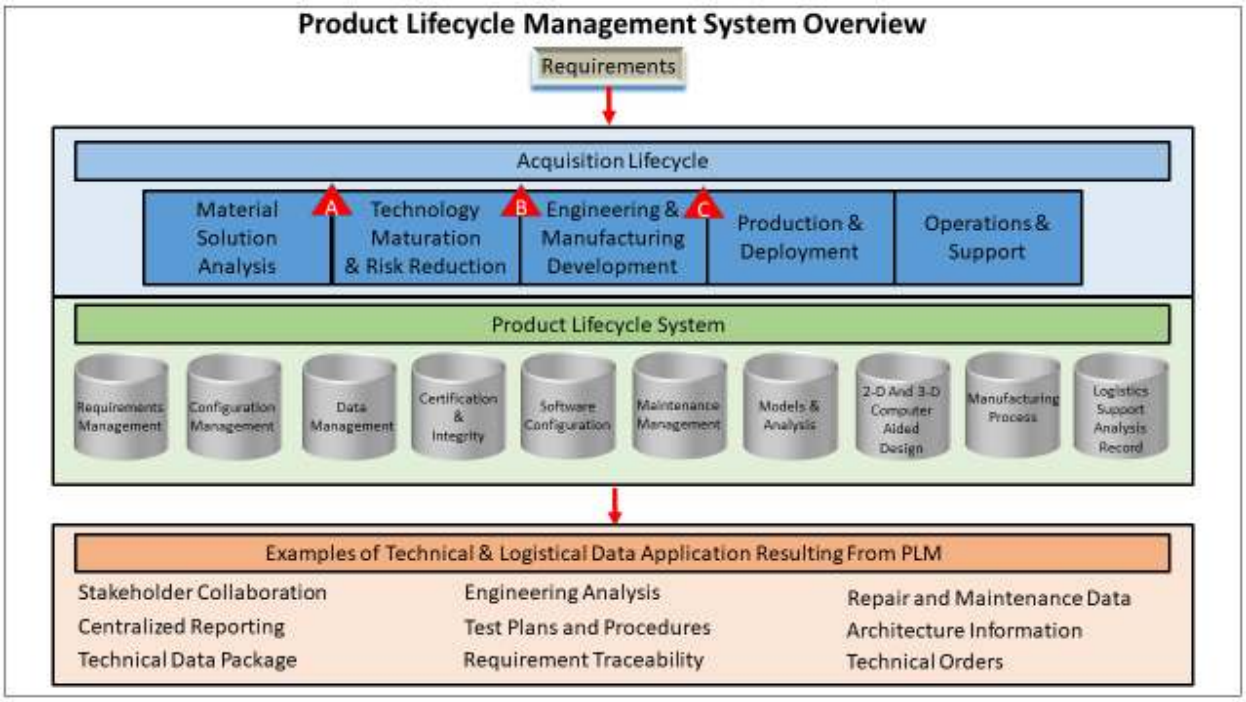


FIGURE 18. Product lifecycle management system overview.

7.2 PLM benefits. Implementing PLM consolidates individual data systems used in the support of lifecycle processes to create a more unified information data structure. It also provides an enterprise data solution to geographically distanced Government and industry programs. This enables a program manager (PM) to work in tandem with program stakeholders in an integrated fashion to enable lifecycle management of weapon systems. Configuration management (CM) and data management roles and processes also benefit by having centralized access to weapon system data related to traceability, engineering, and product support data, along with engineering analysis and decision support. Some benefits associated with PLM include:

a. The ability to maintain the integrity of product data throughout the product's lifecycle

b. Management of universal and secure access to product data

c. The ability to establish and maintain processes used in the creation, modification, maintenance, reporting, and distribution of information

7.3 Defining the DE strategy for PLM. The DE strategy needs to be understood at the enterprise level, as well as at the program acquisition team level. The full PLM needs to be considered and implementation of a DE strategy may be different throughout the product support lifecycle. One product tool that allows DoD organizations to communicate in a DE ecosystem is the high performance computing modernization program (HPCMP). The DoD DE ecosystem and enterprise strategy goals that HPCMP supports are shown on figure 19.

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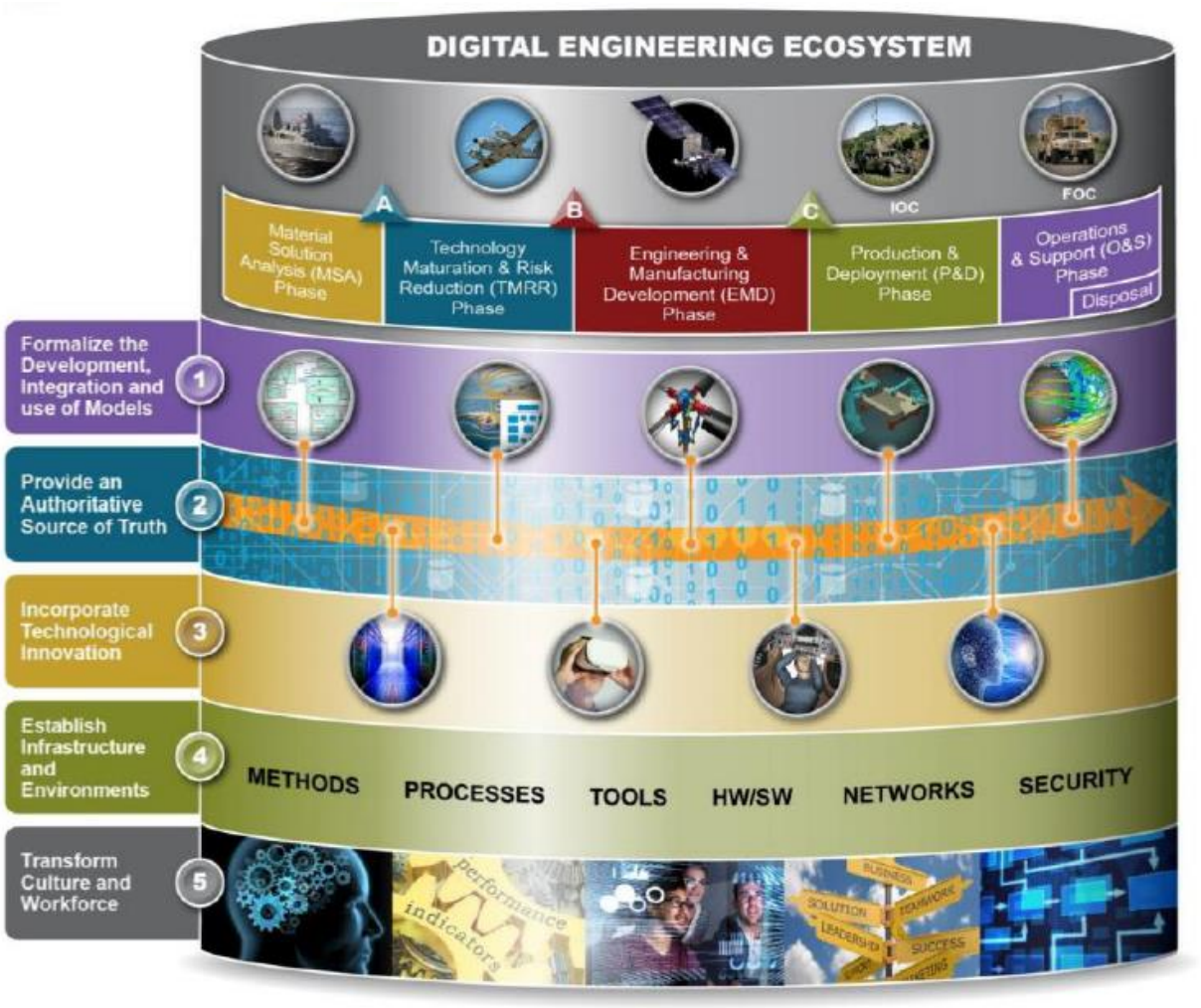


FIGURE 19. DE ecosystem.

7.4 Product support considerations. The consolidated digital transformation involves a modernized logistics data system that includes configuration management, provisioning, readiness modeling, and technical data management support. Incorporation of commercial off the shelf (COTS) capabilities and MOSA should be considered to drive sustainment efficiencies.

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7.5 CM in the DE ecosystem. A digital CM approach with automated tools needs to be established to control and curate product attributes and technical baselines across the total system lifecycle. The DoD CM approach is specified in SAE 649, SAE 649-1, and MIL-HDBK-61. This will enable the lead systems engineer , per the PM's direction, to identify, document, audit, and control schedule, cost, functional, physical, and performance characteristics of the system design. A digital CM process will be established by the configuration manager, per the PM's direction, with resources, staffing roles, and training to allow the automated implementation of all CM functions. Configuration baseline digital threads will be created that include nomenclatures and unique identification labels (as appropriate) for each hardware and software end-product that will maintain proper linkages to their defining data and relevance to the systems design and operational platform. Automated change management activities established will maintain control of hardware and software end-product definition data changes and can trace to all affected items using a digital CCB to properly assess impacts and render approval status. With a digital CM process using models, it allows the ability to capture, record, retrieve, and report status and performance information about the product. This allows the information to be accessible to support other impact analysis and capture the commitments and history of each change. This audit trail facilitates verification testing and analysis of end products for use in other product designs and decisions for enterprise integration efforts. Verification and audit activities are performed with data sets to maintain consistency between the end-product and its configuration information throughout the life cycle. This supports the design integrity and ensures the end-product configuration information is complete, accurate, and current to produce the end-product, along with applicable operation and maintenance instructions, training, and spare and repair parts for release. In the end, this will provide for traceability of mission capability to system requirements to performance and execution metrics.

7.6 DE implementation for PLM and sustainment. The Government has the most direct involvement and the greatest need to be involved during the operations and support (O&S) phase. There is an opportunity to progress to a higher DE stage at the O&S phase. Moving to a higher DE stage needs to be planned out early with each affected prime contractor. Some of the activities that are required during O&S, using a typical aircraft carrier use case example, are shown on figure 20.

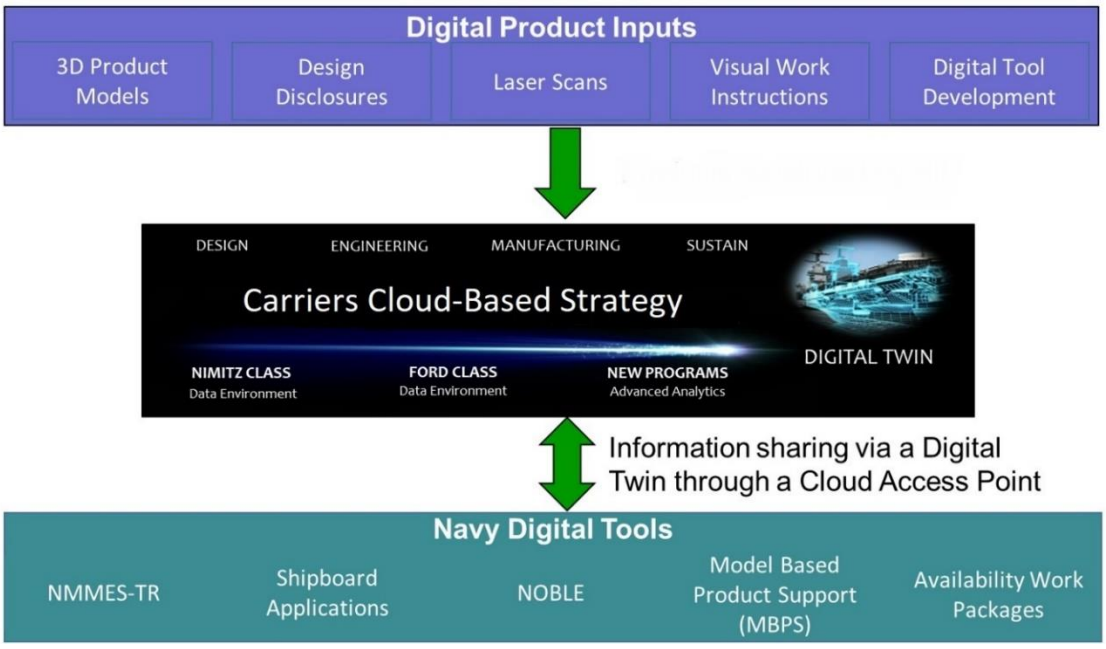


FIGURE 20. Aircraft carrier use case diagram.

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7.7 Logistics DE future state. As DoD adopts DE transformation strategies, the benefits from a modernized logistics portfolio with elevated capabilities should be considered to meet evolving threats. Modern digital solutions that increase effective mission outcomes is needed to transform efforts across the enterprise.

7.8 Logistics DE tools. One current naval tool that integrates the PLM and sustainment activities using an MBSE approach is model-based product support (MBPS). MBPS allows enterprise DE integration that will drive PLM cost reductions. MBPS interfaces are shown on figure 21. Various efforts to support logistics exist within DoD for providing a single source of truth and product data analytics. MBPS was initiated within the Naval Sea Systems Command (NAVSEA); however plans to merge with the United States Marine Corps (USMC) and the Naval Air Systems Command (NAVAIR) aviation product lifecycle management are being pursued. Interfacing with all systems commands and services will also need to be addressed.

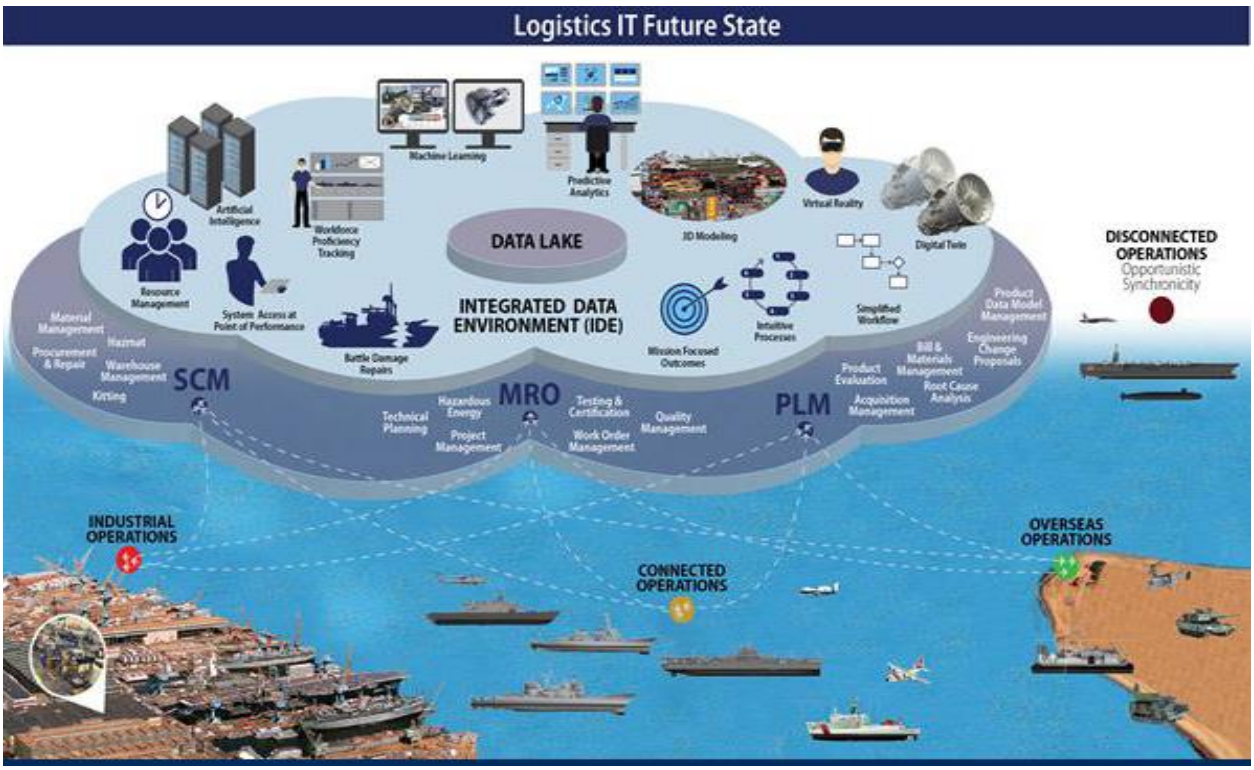


FIGURE 21. MBPS interfaces.

7.9 Product changes within a DE environment. Most DoD acquisition products require product upgrades, tech refresh, repair, and maintenance to ensure that the products are relevant in their current environment and meet operational requirements. An example of managing product changes is shown on figure 22. There is currently no comprehensive, overarching plan, standards, or requirements to incorporate all the developing digital efforts. Without this plan, the process is broken, resulting in the efficiencies paid for not being recognized.

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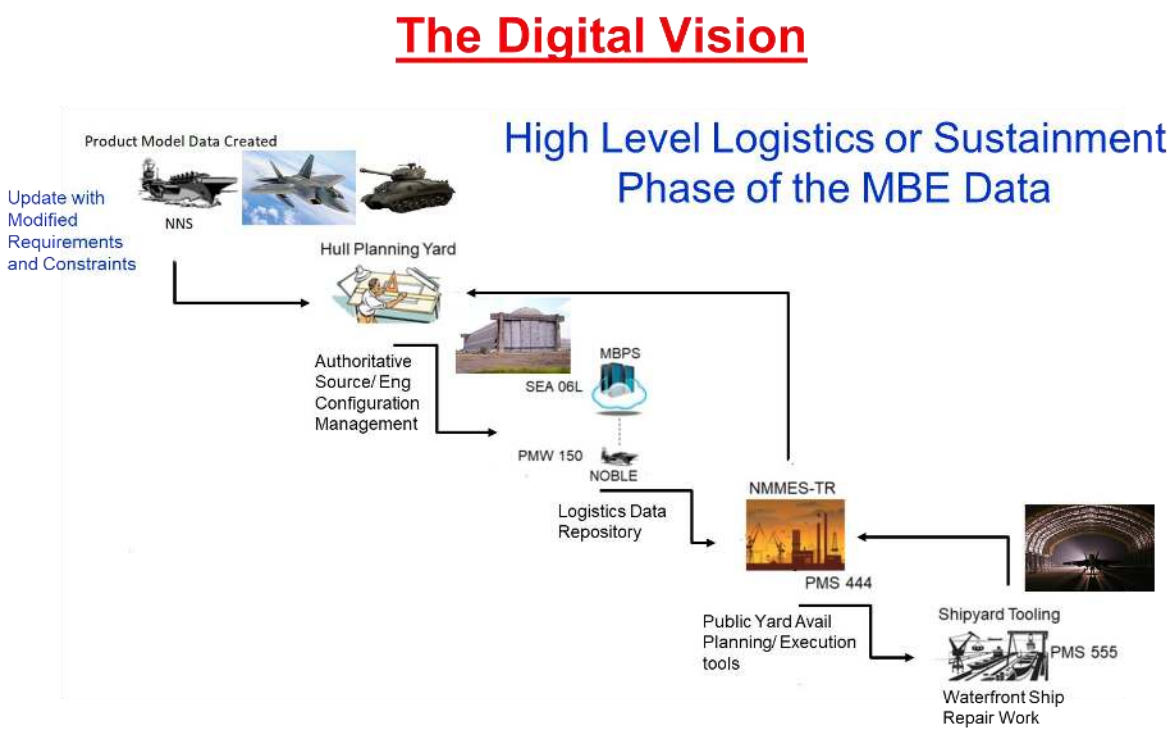


FIGURE 22. Product change management.

7.10 Defense Logistics Agency (DLA) and sustainment support. The DLA is the DoD's supply chain manager for all consumables and provides tailored acquisition support for specified repairables. It is the lead agency in ensuring logistics support during the sustainment phase of weapon system lifecycles.

7.10.1 Digital data in sustainment. Weapon system sustainment requirements for use of digital data should be synchronized with DLA's data management processes and repositories. DLA's technical and quality assurance activities are based on the authoritative data and master data, controlled through the cognitive engineering services. DLA currently has a dual role in support of theDoD:

a. To maintain the consumables supply chains for the Department, across multiple classes of supply

b. To maintain DoD's Joint Additive Manufacturing Model Exchange (JAMMEX), as shown on figure 23.

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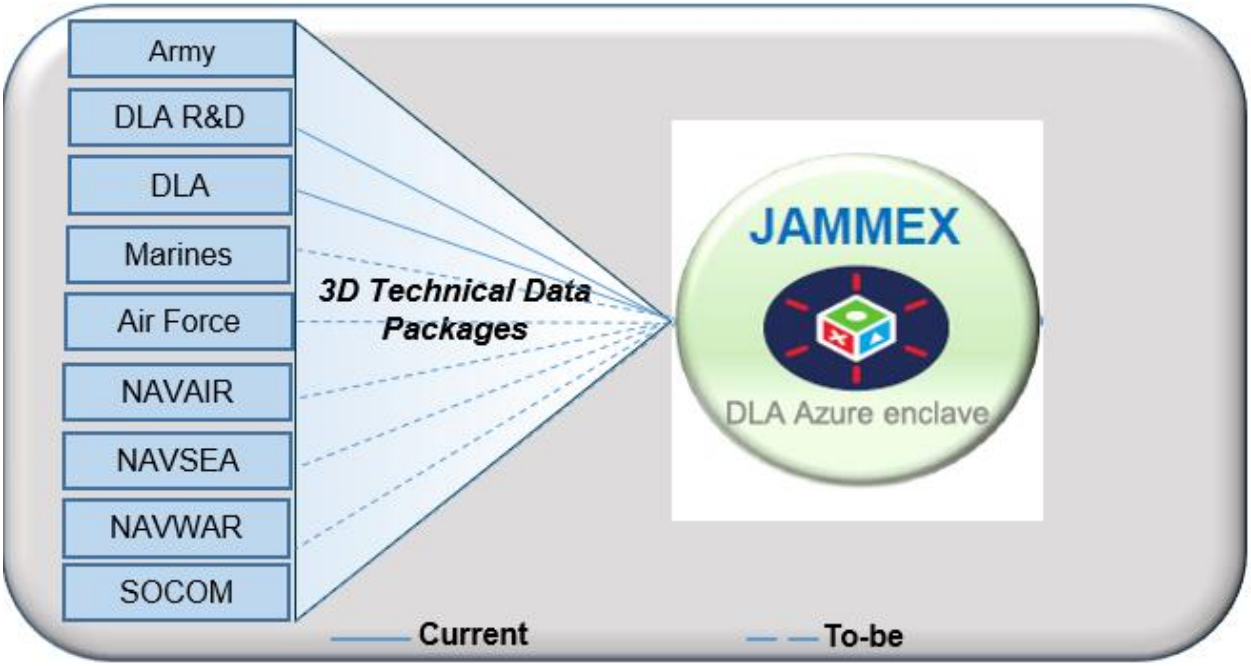


FIGURE 23. Joint additive manufacturing model exchange (JAMMEX).

7.10.2 Systems modernization. DLA is embarking on an enterprise IT modernization that will include expansion of the technical data management system. DLA accesses product data and item performance requirements from the military services' repositories, adding quality assurance and other provisions to the technical requirements in DLA contracts. DLA's IT modernization will expand product data management (PDM) capabilities and repositories to include 3D and other modeling formats. DLA is also exploring digital thread environments and how they may enhance supply chain processes, to include secured sharing of stakeholders' information (customers, DLA, and suppliers) and drive more integrated and timely decisions.

7.10.3 Digital data sharing. As specified in DoDI 5000.93, DLA has been authorized to manage a 3D modelling network and repository, known as JAMMEX, with emphasis on sharing additive manufacturing designs across the DoD. JAMMEX is a secure interservice platform within the DoD that provides capabilities to store, manage, and share 3D models and associated metadata. JAMMEX has been designed to be agnostic and support all file formats.

7.10.3.1 JAMMEX interface with military services' repositories. The JAMMEX system will interface with various military service repositories containing CAD 3D models. The website will allow users to search for and view the 3D models and download the associated file to modify or manufacture the component. Users will also be able to upload new files to JAMMEX where they will be available for search and download systemwide. This capability will give users the critical ability to manufacture parts on an as-needed basis, in locations where resupply is difficult or in situations where time is critical.

7.10.3.2 User access to JAMMEX. The system currently has the following eight repositories: Army, NAVAIR, NAVSEA, Naval Information Warfare Systems Command , USMC, United States Special Operations Command, Air Force, and DLA (additional repositories can be added as required). Users are assigned to the repository for their service branch. Service-designated administrators assign roles and permissions to their users' JAMMEX profiles. These roles and permissions determine what level of access and capability each user has. Generally, a user can view models/metadata and upload/download files. They can also update models/files they have themselves uploaded within their own repository. These permissions can be more or less restrictive depending upon the specific user, based on the repository's security policy and the repository administrator's assessment.

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7.10.3.3 Uploading data into JAMMEX. Models/metadata can be loaded into JAMMEX in a number of ways (e.g., PDF, standard tessellation language [STL], STEP, extensible 3D [x3D]). Data can be loaded from an external database via an automated interface and periodically updated with scheduled jobs. The external database would remain the data administrator and determine what is stored/accessible from JAMMEX. Derivative data can also be loaded to JAMMEX in a one-time dump from an external system and the service will remain the administrator for their data. No further contact with the external system is required. Where no external system/connection is possible (e.g., files are all stored locally), the data can be entered manually, and JAMMEX becomes the sole database administrator for that service's data.

8. CONTRACTING FOR DE

8.1 Contracting language for DE implementation. Enforceable contract language is critical to successful DE. Each request for proposal (RFP) should be properly tailored to acquire sufficient IP deliverables and associated IP license rights to implement DE (which includes implementing MOSA). Accordingly, the following discussion should be considered a point of departure regarding how an RFP should address these topics consistent with the program's capability development document , acquisition strategy, SEP, and lifecycle sustainment plan. The appendices within this handbook include sample contracting language that programs can use to complete their data delivery documentation. Figure 24 shows how data exchange is defined and controlled using contract language within an IDE, using multi-level classification.

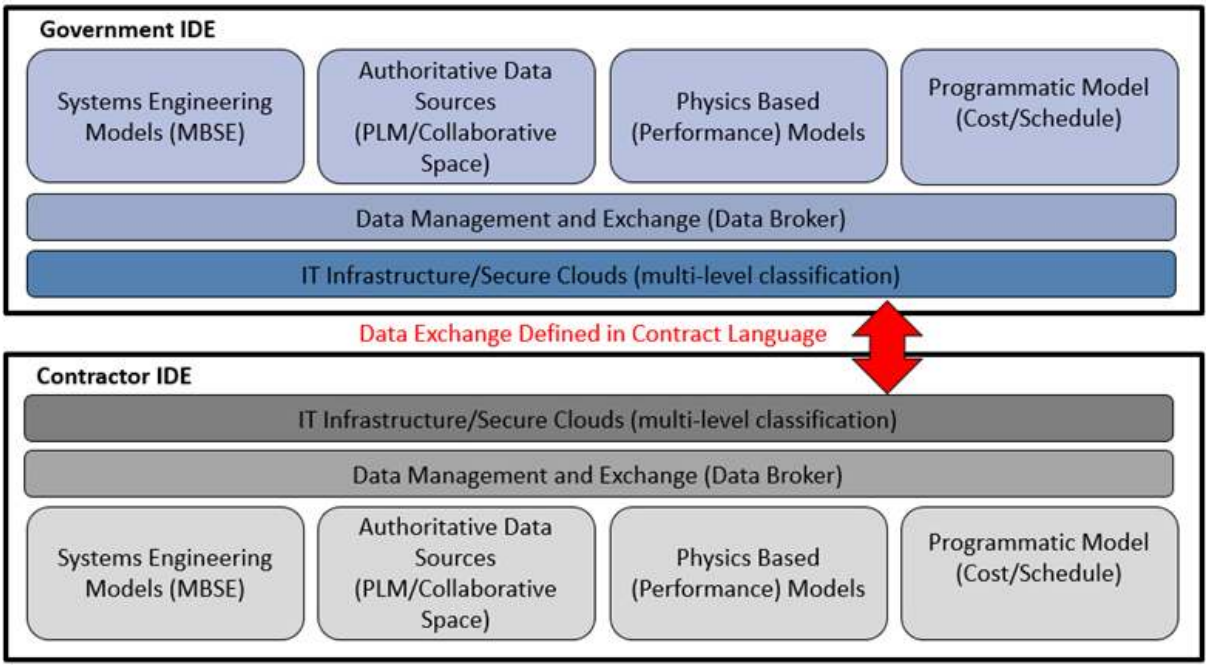


FIGURE 24. Government and contractor IDE interaction example.

8.2 Sample CDRL and TDP option selection worksheet for 3D data delivery. See Appendix A for a sample CDRL or DD Form 1423 and the TDP option selection worksheet for contract data delivery. The TDP (as specified in MIL-STD-31000) will be further managed by the program office configuration management master plan and the prime contractor configuration management plan.

8.3 Sample contract clause for electronic delivery of CDRLs. See Appendix B for a sample contract clause with prioritized list of data formats for electronic delivery.

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8.4 Sample one-time DID. See Appendix C for a sample one-time DID. Due to the complexities of specific 3D product model data attributes, a general DID may not be effective. The tailored one-time DID is referenced by the CDRL and will contain the specific data requirements. Another option to avoid a complex one-time DID would be to include the DID requirements within the contract. The US Army uses this option to avoid the one-time DID process.

8.5 Sample TDP drawing metadata format. See Appendix D for sample TDP drawing metadata formats that will be associated with the contract DID or one-time DID.

8.6 Sample functional groupings of drawings. See Appendix E for a sample listing of drawing functional groupings.

8.7 Sample values for associated lists. See Appendix F for a list of valid values for associated lists, as specified in ASME Y14.34.

8.8 Sample values for data rights categories. See Appendix G for a list of valid values for data rights categories.

9. CYBERSECURITY REQUIREMENTS

9.1 Cybersecurity planning. As specified in DoDI 5000.02 and DoDI 8500.01, in order to protect the program's data, cybersecurity requirements need to be established early in the acquisition process and fully integrated throughout the lifecycle. Because cybersecurity can be costly to visualize and test, MBSE is the ideal methodology for representing and analyzing the cybersecurity of a system. The full details and timeline for integrating cybersecurity into a program's lifecycle can be found in the DoD Program Manager Guidebook for Integrating the Cybersecurity Risk Management Framework (RMF) into the System Acquisition Lifecycle. Rather than duplicating information found in this and other documents, this handbook will outline cybersecurity procedures unique to DE.

9.1.1 Cybersecurity professionals. Cybersecurity professionals are an essential part of the systems engineering design team and should be included early in the development of a new capability. While the MBSE model may be built under the direction of systems engineers, cybersecurity professionals will lead the development of system security within the MBSE environment. All design decisions pertaining to cybersecurity should be made in consultation with the cybersecurity professionals, including the program's RMF activities of security categorization, security control selection and implementation, security control assessment, information system authorization, and security control monitoring, as well as system security engineering tasks such as cybersecurity risk assessments, tabletop events, and test procedures

9.1.2 DE development environment. Of particular note is the need for correct Government and contractor assessment of the DE development environment. Because DE aggregates data that has previously been disparate, system security engineers (SSE) are consulted about the digital environment in which the system will be developed. A data classification study to include inference and aggregation analysis is conducted for all systems, not just for those with multi-security classification levels. SSEs should use guidance for classifying databases, rather than documents, when determining the appropriate development environment and access controls for DE artifacts.

9.1.3 Modeling integrated cybersecurity design. Whenever possible, cybersecurity modeling and analysis should be integrated into system design models alongside other system design requirements and decisions. Information exchange requirements , RMF technical controls, and cybersecurity-related key performance parameters and system functions should be modeled in the same fashion, and in the same model, as other system requirements and functions. This allows maximal sharing between cybersecurity subject matter experts and other engineers.

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9.1.4 Modeling cyber resilience. Cyber resilience is the system's ability to continuously deliver the capability and meet mission requirements, despite hostile cyber operations. The capability will need to address operation in degraded mode while supporting missions. Modeling and simulation can examine degraded operations and provide insight to recovery path options. Some areas of modeling resilience may include modeling of the components of zero trust architecture, modeling policy engines and trust algorithms, and modeling system responses to attack paths such as honeypot activities or hot and cold redundancy.

9.1.5 Cyber threat modeling. Threat modeling analyzes what effect a threat has on our system capabilities. Design should be focused upon reducing and eliminating risk elements as much as it is on developing capabilities. DE modeling and simulation activities can use threat modeling to improve design decisions. The selected modeling tools should permit quick add, remove, and movement operations of architectural components as more is known about the threat to optimize the technical solution. The context of each threat modeling activity can use known tactics, techniques, and procedures (TTP) to uncover unique threat vectors where safeguards will improve security posture. Cyber-actor strategies can be modeled and walked through in set phases. Each phase will allow the design team to focus on specific TTP sets. Output from cybersecurity design provides inputs to the requirements verification traceability matrix (RVTM), system requirements specification, and system/subsystem specification. Some areas of threat modeling include identifying areas of initial exploitation, persistent access points, malicious tool installation, and adversary lateral movement within the system.

9.2 System modeling guidance. The following section provides further guidance on the scope of creating an MBSE model for cybersecurity engineering:

9.2.1 System architectural views. Architectural views provide the systems operational concept and key parameters for analysis of cybersecurity. As specified in the DoDAF , legacy systems can continue to leverage system views but should transition to SvcV when their architecture is updated. New capabilities like zero trust architecture and new systems design should use SvcV. The operational overview (operational viewpoint [OV]-1), connectivity diagrams, and resource flow tables (OV-3, SvcV-2, and SvcV-6) provide base information that will assist with threat modeling. The OV-1 is a high-level graphical illustration of the operational concept. The OV-3 identifies the resource transfers to achieve a specific operational task for expected users. The SvcV-6 specifies the characteristics of the services resource flows exchanged between systems and system boundaries. The SvcV-2 visually demonstrates the data exchanged across the system boundary and can be used for system boundary identification for RMF. A redacted example of SvcV-2 is illustrated on figure 25. Ensuring applicable cybersecurity controls and standards are integrated into the functional baseline architecture will advance precise standardized security postures ready to defend the capabilities fielded. RVTMs demonstrate how cybersecurity requirements are allocated to, and satisfied by, system components. Mission traceability tables identify desired effects, operational and system tasks associated with those effects, and system parts that play a role in the effects, thus, identifying mission essential functions and mission criticality of system components. Implementing RMF and standardization early in the systems engineering process will ensure cybersecurity is accounted for during systems requirements review and in the system functional review.

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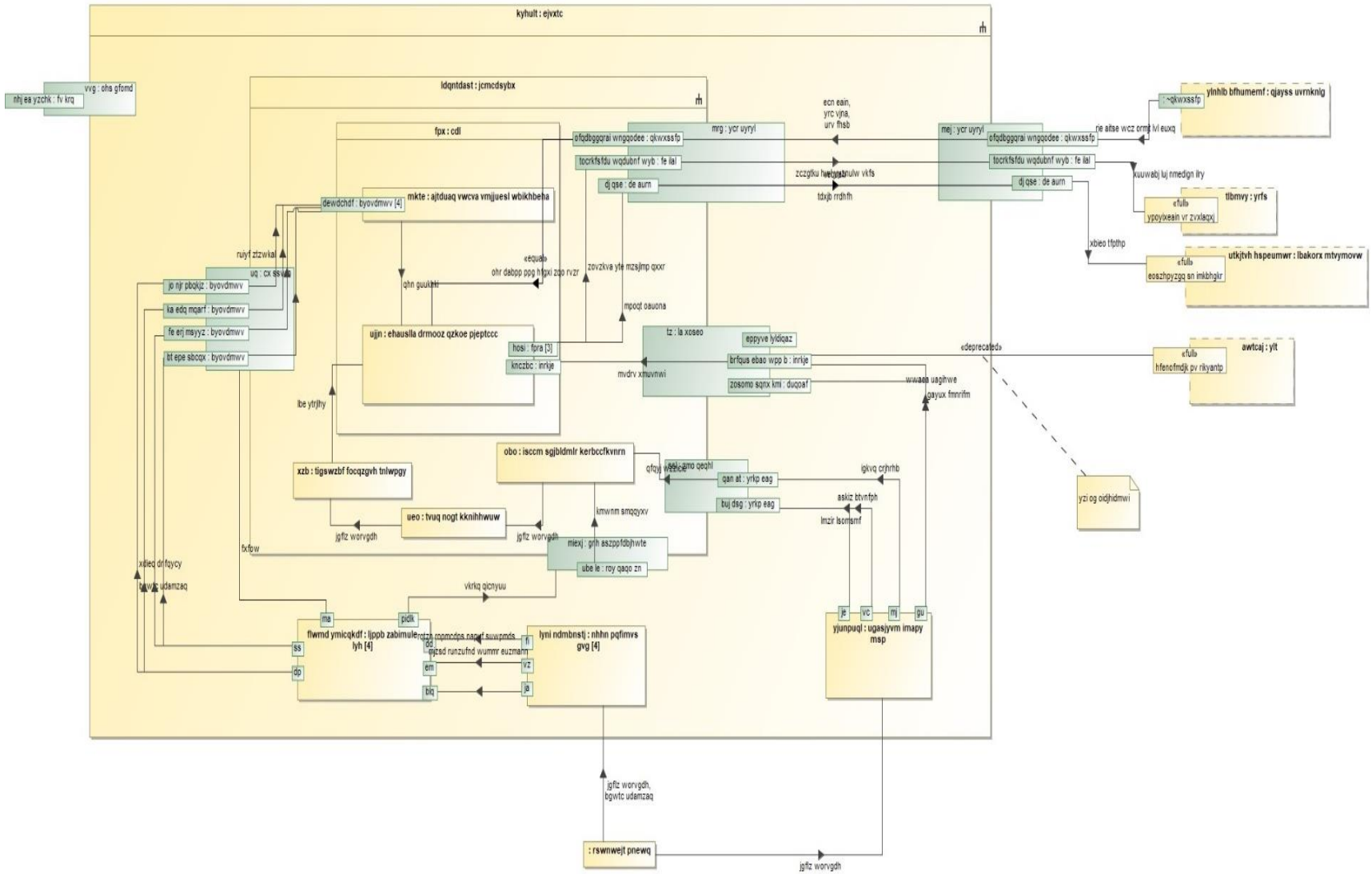


FIGURE 25. Connectivity diagram example.

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9.2.2 MBSE model content for cybersecurity. As a starting point, it is recommended that MBSE system models capture the following data for cybersecurity assessment:

a. System, subsystem, or components:

(1) Hardware, firmware, and software (as known, include version numbers to cross-reference with common vulnerability enumerations)

(2) System states and functions, including maintenance states/functions and degraded states

(3) Data flows between system components

(4) System classification (as distinct from MBSE model classification), export status, controlled unclassified information and controlled technical information

(5) System operating environment, system boundary, and external systems to which the system is connected

(6) Manufacturer, model number, and update number

(7) Access control methods

(8) Authentication methods (where applicable)

(9) Supply chain risk

(10) Data encryption/decryption functions and type of encryption

(11) Cyberspace protection conditions behavioral state changes, where applicable

(12) Vulnerability scanning procedures or products

(13) Internet protocol and media access control addresses, where applicable

(14) Mission criticality of system or system part

(15) Ports and other system interfaces

(16) Protocols and services, where applicable

(17) Maximum latency, where applicable

b. Data elements flowed within system:

(1) Data classification

c. System users:

(1) Security clearance level

9.3 MBSE cybersecurity modeling standards for services. It is strongly recommended that each service standardize the model formats and views required for RMF using MBSE schemas, profiles, shared model libraries, and style guides. The RMF MBSE modeling standards should be held at the service level to be aligned with the level of authority of the authorizing office. Because many RMF mitigations are inherited from systems external to the system under evaluation, it is important that the modeling pattern is standardized to ensure that all systems understand the risk they are inheriting, and that models can be read together to understand the whole picture. At a minimum, RMF controls and control correlation identifiers should be created as a shared MBSE model to avoid rework across MBSE models. Cybersecurity modeling standards for services should account for the MBSE model content captured in 9.2.2.

9.3.1 RMF officials. RMF officials should consult heavily with their service's broader MBSE standards team and vendors to ensure that the RMF modeling format standards are compatible with larger systems engineering modeling format standards. RMF qualified validators, information systems security managers , and information systems security engineers should also be consulted to determine that the correct data is captured within the MBSE model. Once the RMF MBSE modeling standard has been piloted, the service should begin training all RMF qualified validators in reading MBSE models to understand the data contained therein. All service ISSMs, ISSEs, and information system security officers may be trained in how to enter RMF data into the MBSE system model.

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9.3.2 Service cybersecurity. Services may wish to develop additional MBSE modeling standards for other cybersecurity data, such as cyber survivability attributes, cybersecurity risk assessments, cybersecurity tabletop events, or program protection plan tables. Standardizing the formatting of this data can reduce errors in system vulnerability assessments, particularly with integrated systems, and accelerate workforce adoption of DE practices for cybersecurity. The service MBSE standards should be required in MBSE contracts to reduce model rework.

9.4 Automated cybersecurity assessments. All cybersecurity assessments should be completed independently of the MBSE system model (for instance, in an external tool or in an MBSE analysis model external to the system model). This will help a program avoid overclassification of their MBSE system models, while also protecting system data by keeping need-to-know data out of the system model shared by a larger audience. However, cybersecurity assessments should use the MBSE model as the primary or sole source of information in order to retain a single source of truth. There are many DoD hardware and software tools in development that can assist with automated cybersecurity assessment of a system by using system scans or MBSE system models. Because many of these tools produce different types of risk assessment, it is recommended that each service perform an assessment of alternatives to determine which tools best capture system risk to meet the service's needs. When assessing an automated cybersecurity assessment tool that ingests MBSE system models, each service should also measure the tool against service MBSE modeling format standards to determine the rework necessary to use the tool. Rework of the MBSE systems engineering model to accommodate an assessment tool should be minimized, and services are encouraged to negotiate with tool vendors about reading data encoded by the service's MBSE standards.

10. NOTES

10.1 Intended use. This handbook is intended to provide those in the DOD acquisition arena and supporting contractors with a characterization of an MBE and guidelines on how it is achieved.

10.2 Subject term (key word) listing.

Digital Thread

Digital Twin

Model-based Enterprise (MBE)

Model-based Definition (MBD)

Model-based System Engineering (MBSE)

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APPENDIX A. CONTACT DATA REQUIREMENTS LIST (CDRL) FOR 3D DATA DELIVERY

A.1 SCOPE

A.1.1 Scope. The CDRL provides a centralized list of authorized data requirements a contract should provide when delivering data. A sample CDRL that can be used by a program office is included in A.2 and a TDP option selection worksheet is included in A.3.

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A.2 SAMPLE CDRL

CONTRACT DATA REQUIREMENTS LIST *(1 Data Item)*	*Form Approved* *OMB No. 0704-0188*

Public reporting for this collection of information is estimated to average 110 hours per response including the time for reviewing instructions, searching existing data sources, gathering and maintaining the data needed, and completing and reviewing the collection of information. Send comments regarding this burden estimate or any other aspect of this collection of information including suggestions for reducing this burden to Department of Defense, Washington Headquarters Services, Directorate for Information Operations and Reports, 1215 Jefferson Davis Highway, Suite 1204, Arlington, VA 22202-4302, and to the Office of Management and Budget, Paperwork Reduction Project (0704-0188), Washington, DC 20503. Please DO NOT RETURN your form to either of these addresses. Send completed form to the Government Issuing Contracting Officer for the Contract/PR No. listed in Block E.

A. CONTRACT LINE ITEM NO. XXXX	**B. EXHIBIT** A	**C. CATEGORY:** TDP _X__ TM ___ OTHER
D. SYSTEM/ITEM DoD Acquisition Example	**E. CONTRACT/PR NO.** N000XX-22-N-XXXX	**F. CONTRACTOR** Prime Contractor

1. DATA ITEM NO. A00X	2. TITLE OF DATA ITEM Product Drawings/Models and Associated Lists	3. SUBTITLE Solid Model Data – DoD System
4. AUTHORITY (Data Acquisition Document No.) OT-XXXX-XXXXX	5. CONTRACT REFERENCE Section C, Part X	6. REQURING OFFICE DoD Program Office

7. DD 250 REQ DD	9. DIST STATEMENT REQUIRED "D" See Block 16	10. FREQUENCY ASREQ	12. DATE OF FIRST SUBMISSION See Block 16	14, DISTRIBUTION
8. APP CODE A		11. AS OF DATE N/A	13. DATE OF SUBSEQUENT SUBMISSION See Block 16	

16. REMARKS

Block 9 – Use the following*: (Insert appropriate distribution statement, export warning, and destruction notice in accordance with DoD Directive 5230.24 or DD Form 254.)*

Blocks 12 & 13 – No later than 60 days after contract award the contractor/developer shall submit native electronic Solid Model Data to the designated Cloud environment. Product drawings (assembly, control, detail, etc.) and lists (parts, data, and index) will be furnished in the final drawing package. The Government will review the submitted data to ensure compliance with contractual requirements. All other drawings developed under this contract shall adhere to the format approved by the Government.

In-process technical review: The contractor/developer shall submit a copy of the completed drawing package 75 days before the PCA. When the contractor/developer has received comments resulting from the Government in-process technical review, the contractor/developer shall review the Government comments, make corrections to the entire package and resubmit all new and revised drawings within 30 days of comment receipt. Government comments, whether directed against a specific drawing, or made as a general comment, may apply to other drawings that have not been reviewed by the Government. The Government will subject resubmitted drawings and other drawings not previously reviewed to another technical review. Any additional comments generated as a result of subsequent technical reviews shall also be furnished to the contractor/developer for action. This process shall be repeated as necessary. Response time for each submittal thereafter shall be 15 days.

Final delivery: The contractor/developer shall submit the final drawing package within 30 days of Government request.

Block 14 – Data shall be delivered as native Solid Model Data or in digital format in compliance with the attached TDP Option Selection Worksheet. Native Solid Model Data delivered to the designated Cloud environment shall be considered as the data item submittal to Addressees. Additional product drawings and lists delivered to the designated electronic repository shall be considered as the data item submitted to Addressees. Addressee notification of data availability shall be accomplished through an automated notification.

Final submittals shall be by DD Form 250, as indicated in Block 7.

a. ADDRESSEE	b. COPIES		
	Draft	Final	
		Reg	Repro
In-Process Format Review:			
DoD Program Office (Code XXX)	0	1	0
DoD Acquisition Agent (Code XXX)	1	1	0
DoD ISEA (Code XXX)	0	1	0
DoD Repository (Code XXX)	0	1	0
Other DoD Activity (Code XXX)	0	1	0
See Block 16			
15. TOTAL	1	5	0

G. PREPARED BY	**H. DATE**	**I. APPROVED BY**	**J. DATE**

DD Form 1423-1, JUN 90 (EG) *Previous editions are obsolete.* Page __ of __ Pages

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A.3 SAMPLE TDP OPTION SELECTION WORKSHEET

TDP OPTION SELECTION WORKSHEET			
SYSTEM: Sample Navy Program Contract Example			DATE PREPARED:
A.CONTRACT NO. As Applicable	B. EXHIBIT/ATTACHMENT As Applicable	C. CLIN As Applicable	D. CDRL DATA ITEM NO. As Applicable

1.TDP Lifecycle Level (X and complete as applicable.)	
A. ☐ CONCEPTUAL LEVEL ☐ DEVELOPMENTAL LEVEL ☒ PRODUCTION LEVEL	B. REMARKS:

2.TYPE AND FORMAT (X all that apply and complete as applicable)	
A. ☒ TYPE 2D: 2D DRAWINGS ☐ TYPE 3D: 3D MODEL ONLY ☒ TYPE 3D: 3D MODELS WITH ASSOCIATED 2D DRAWINGS See Block 9	B. ☒ NATIVE CAD (Native File format the data was created in. Specify level of annotation.): ☐ ISO 10303 STEP FORMAT (SPECIFY): ☒ ISO 32000 PORTABLE DOCUMENT FORMAT (PDF) ☒ OTHER ELECTRONIC FORMAT (SPECIFY TYPE) MODEL ORGANIZATIONAL SCHEMA IAW ASME Y14.47 (Specify Appendix B) NEUTRAL CAD FORMAT (Specify e.g. ISO 10303, AP203, AP214, AP216, or AP242) (Specify e.g. 3D PDF, JT) PDF required for 2D; 3D PDF format for full MBD – Model only. (Specify Microsoft EXCEL, CSV, or XML) ☒ HARDCOPY REMARKS: Cloud environment defined in contract

3. CAGE Code AND DOCUMENT NUMBERS	A. ☒ CONTRACTOR CAGE AND DOCUMENT NUMBERS – See Block 9 ☒ GOVERNMENT CAGE (COMPLETE 3B & 3C OR 3D) – See Block 9	D. To Be Assigned By: PMA-XXX As specified in the contract
B. USE CAGE CODE: 30003	C. USE DOCUMENT NUMBERS: See Block 9	

4. DRAWING FORMATS (X one and complete as applicable)
☒ CONTRACTOR FORMAT. ☐ GOVERNMENT FORMAT. REMARKS: _____ CONTRACTOR CAGE CODES SHALL BE USED FOR COMPANY STANDARDS

5. TDP ELEMENTS REQUIRED (X all that apply)
☐ ELEMENTS REQUIRED TO BE DETERMINED BY CONTRACTOR or THE FOLLOWING ARE REQUIRED: ☐ CONCEPTUAL DRAWINGS/MODELS AND ASSOCIATED LISTS ☐ DEVELOPMENTAL DESIGN DRAWINGS/MODELS AND ASSOCIATED LISTS ☒ PRODUCT DRAWINGS/MODELS AND ASSOCIATED LIST – CDRL XXXX ☐ COMMERCIAL DRAWINGS/MODELS AND ASSOCIATED LIST

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☐ QUALITY ASSURANCE PROVISIONS

☒ SPECIAL INSPECTION EQUIPMENT (SIE) DRAWINGS/MODELS AND ASSOCIATED LISTS – CDRL XXXX

☐ SPECIFICATIONS

☒ SOFTWARE DOCUMENTATION – CDRL XXXX

☒ QUALITY ASSURANCE PROVISIONS / METADATA / SUPPLEMENTARY TECHNICAL DATA – CDRL XXXX
For all items requiring Acceptance Testing (DoD-STD-2101).
SOURCE CONTROL DRAWING (SOCD) APPROVAL REQUEST – CDRL XXXX

6. ASSOCIATED LISTS (X and complete as applicable)

☒	A. PARTS LIST (X ONE)	☐ (1) INTEGRAL	☒ (2) REQUIRED (CONTRACTOR SELECT)
☒	B. DATA LIST (X ONE)	☐ (1) NOT REQUIRED	☒ (2) REQUIRED (Lowest level of repair/indenture)
☒	C. INDEX LIST (X ONE)	☐ (1) NOT REQUIRED	☒ (2) REQUIRED (Lowest level of repair/indenture)
☒	D. WIRING LIST (X ONE)	☐ (1) NOT REQUIRED	☒ (2) REQUIRED (Lowest level of repair/indenture)
☒	E. INDENTURE DATA LIST (X ONE)	☒ (1) NOT REQUIRED	☐ (2) REQUIRED (SPECIFY LEVELS OF ASSEMBLY)
☒	F. APPLICATION LISTS (X ONE)	☐ (1) NOT REQUIRED	☒ (2) REQUIRED (CONTRACTOR SELECT)

7. APPLICABLILTY OF STANDARDS. The following Standards apply: (X as applicable)

☒ ASME Y14.100 ENGINEERING DRAWING PRACTICES WITH APPENDICES	☒ ASME Y14.24 TYPES AND APPLICATIONS OF ENGINEERING DRAWINGS	☐ OTHER STANDARDS APPLY AS DESCRIBED:
	☒ ASME Y14.34 ASSOCIATED LIST	
	☒ ASME Y14.35 REVISION OF ENGINEERING DRAWINGS AND ASSOCIATED LIST	COMPANY STANDARDS PERMITTED ☐ YES ☒ NO
☒ B ☒ C ☒ D ☒ E	☒ ASME Y14.41 DIGITAL PRODUCT DATA PRACTICES	
	☒ ASME Y14.5 DIMENSIONING AND TOLERANCES	

8. OTHER TAILORING

Blocks 2A and 2B – The data delivery format requirements are as follows and shall be used by the contractor as required for the 2D and 3D data being delivered as part of the TDP:

Block 2A – 2D drawings – will be delivered in their native file format and with ISO 32000 PDF files for non-CAD users. If native file format is not available, then the current source file format and PDF will be submitted to the government.

Block 2B – 3D Models "only" – or full Model Based Definition (MBD) submittals will be fully annotated such that the model serves as the master source data for all geometry, dimensions, tolerances, notes, annotations, attributes, and product and manufacturing information (PMI) necessary to describe a product. 3D Model only submittals will be delivered in native CAD format and will include a validated STEP file and a viewable format such as 3D PDF or ISO 14739 PRC compliant 3D viewable. When delivering 3D Model only submittals a Model Organizational Schema IAW ASME Y14.47 shall be included with the delivery to ensure the government has complete understanding of how the models are structured and organized.

3D Models with Associated 2D Drawings – 3D Models used to generate 2D drawings will be submitted in native CAD format. The native CAD files will be a complete submittal with model and drawing files. The data delivery will include ISO 32000 PDF files for 2D drawings.

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Block 2C – TDP Metadata provides the administrative data required for TDP management. Refer to CDRL XXXX TDP Metadata and One-Time DID XXXX for specific metadata format requirements. Metadata to be delivered in Microsoft Excel file format, CSV format, or XML format. Coordinate with PMA-XXX for valid field values to be used. TDP Metadata will be provided for the following TDP Elements checked in Block 5: Product Drawings/Models and Associated Lists (CDRL XXXX) Special Inspection Equipment Drawings, Models, and Associated Lists (CDRL XXXX) pecial Tooling Drawings, Models, and Associated Lists (CDRL XXXX)

Block 2D – Contractor required to provide hardcopies of Associated Lists to support In-Process reviews and Physical Configuration Audits (PCAs) as requested by the Government. Not required for Final TDP Delivery.

Block 2E – Supplemental Technical Data shall include engineering analysis to include finite element models and finite element analysis (e.g., airframe loads models, wing analysis/geometric analysis/load sets, internal fatigue models, damage tolerance analysis, strength/fatigue analysis, engineering reports, and modeling and simulation data). Include embedded metadata to the model file.

Blocks 3A, 3B, 3C, 3D – Government CAGE and Drawing numbers required for all new drawings generated as part of this effort. Existing drawings, or revisions to existing drawings may retain Contractor drawing number and CAGE except when a new item identification is being established. A new NAVSEA drawing, model, or associated list will be created when creating new item identification (e.g., when creating a new “dash” configuration, or when identifying a new item). Contact Program XXX, for issuance of NAVSEA engineering drawing numbers (XXXAS0100-XXXXAS9999). Contractor is required to maintain tracking of NAVSEA engineering drawing number usage.

Block 8 – Company standards shall not be referenced in NAVSEA drawings, models, associated lists and other TDP products. If a company standard is referenced in a contractor drawing, model, or associated list that is being delivered as part of the government TDP, a copy of the company standard will be delivered with the final TDP delivery in order to meet design disclosure requirements.

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APPENDIX B. ELECTRONIC DELIVERY OF CDRLS

B.1 SCOPE

B.1.1 Scope. This appendix is an example of a contract clause. It recommends specific data formats in which TDP documents (drawings, specifications, and reports) and data on procedures, systems, and applications should be delivered. The contract can also specify preferences with data formats to drive data exchange efficiencies.

B.2 SAMPLE REQUIREMENTS FOR DRAWINGS AND 3D MODELS

ELECTRONIC DELIVERY OF

CONTRACT DATA REQUIREMENTS LISTS (CDRLs)

The following shall be incorporated into the Contract Data Requirements List (CDRL) Special Instructions. Unique requirements that differ from the information defined below shall be specified in the contract.

1. Drawings shall be delivered in accordance with:

   a. Viewing orientation may be 0, 90, 180 or 270.
   b. Image sizes, including over scanning shall be per MIL-PRF-28002 for drawing sizes "A" through "F" and "A4" through "A0" as defined in ASME Y14.1.
   c. Drawings are reviewed in native or PDF formats, but delivered in native format.
   d. Preferred Solid Model Data native formats to avoid data translation are; Creo 5.0 and above, CATIA V4 or above, Siemens NX/Unigraphics NX, SolidWorks, Autodesk Inventor. These platforms allow improved productively in a Multi-CAD environment.

2. Other Technical Data Package (TDP) documents (i.e. "A" size drawings, specifications and reports) and other data items shall be delivered on electronic media (via a World Wide Web page, or electronic mail as agreed to per individual contract. Windows or Macintosh format to be identified for each contract. Use of software applications other than those listed below are to be coordinated between Government and Contractor prior to delivery.

   a. All text shall be in Microsoft Word and/or PDF format.
   b. All graphs and charts shall be in Microsoft Project/Power Point.
   c. Microsoft Excel may be used where applicable.
   d. Microsoft Power Point/Project may be used where applicable.
   e. When the above listed software programs do not meet the formatting required for a particular application, the software program shall be saved in a Microsoft Word/Microsoft Office document in a readable and correctable format.

   f. Where the text is a part of the document, Microsoft Word shall be the preferred deliverable, although PDF formats are acceptable. A document shall not require further assembly or manipulation prior to print and distribution.

3. Solid Model Data or 3D Computer-Aided Design (CAD) Models, when required, shall be delivered in accordance with the following order of preference:

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a. Native 3D CAD data files shall be used as the primary method of data file transfer using a preferred Solid Model Data native data format.
b. If a preferred Solid Model Data native data format is unavailable, the ISO 10303, Standard for the Exchange of Product Model Data (STEP), Application Protocol (AP) Part(s) 242 (See *NOTE) which is compatible with Government and Contractor systems and applications shall be used if native data is unavailable.
c. If native data or compatibility with STEP is unavailable, an IGES file which is compatible with Government and Contractor systems and applications and has been shown to be effective in the translation of data files through test shall be used. Data files will be created using MIL-PRF-28000, Class II as guidance.

4. When Government and Contractor procedures, systems and applications are data files may be delivered using an established on-line Continuous Acquisition and Life Cycle Support (CALS)/Contractor Integrated Technical Information System (CITIS) approach.

*NOTE: ISO 10303-XXX AP: Application Protocol: Documents that specify the format for representing product data within a set of related processes or activities that can be added by contract.

AP210: AP for Electronic Assembly, Interconnect and Packaging Design
AP215: AP for Ship Arrangement
AP216: AP for Ship Moulded Forms
AP218: AP for Ship Structure
AP227ed2: AP for Plant Spatial Configuration
AP239: AP for Product Life Cycle Support
AP242: AP for Managed Model Based 3D Engineering

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APPENDIX C. SAMPLE ONE-TIME DID

C.1 SCOPE

C.1.1 Scope. The one-time DID provides specific data format requirements. Most program offices will need a customized approach using a one-time DID. The DID should include a standardized list of data used with interfacing subsystems and systems, as well as program specific data requirements.

C.2 SAMPLE ONE-TIME DID

Title: TECHNICAL DATA PACKAGE (TDP) METADATA REQUIREMENTS
Number: OT-1234-56789 **Approval Date:** XXXXXXXXX

AMSC Number: *N/A* *Limitation:*

DTIC Applicable: No **GIDEP Applicable:**

Preparing Activity: **Project Number:**

Applicable Forms:

Use, Relationships: Technical Data Package (TDP) Metadata is used to store, manage, manipulate and provide access to TDP elements stored in the Program Authoritative Source repository. The metadata required is the data needed by the information management system, used by the Procuring Activity to perform these functions. The metadata needs to be delivered in a format that the information management system can process. TDP Metadata includes all relations, parameters and system information used in managing the TDP.

This DID contains the format, content, and intended use information for the data product resulting from the work task described in the Statement of Work (SOW). This DID is for one-time use for Contract number (*insert Contract Number for TDP Procurement).*

Requirements:

1. References: An example of the TDP Metadata format can be obtained from the following address: (*insert the Program address and Point of Contact with phone number – should be the TDP Life Cycle Management (LCM) for the Program).*

2. TDP Metadata shall be delivered with all TDP data file deliveries and the delivery schedule for TDP Metadata shall coincide with the delivery requirements of the appropriate TDP Element Data delivery.

3. Format: The TDP Metadata shall be submitted in a comma separated value (csv), XML, or Microsoft Excel file format. Required data elements and metadata format requirements are detailed below. The TDP Metadata deliverable is separated into 3 tables: TDP Drawings, TDP Parts, and TDP Valid value types:

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APPENDIX D. SAMPLE TDP DRAWING METADATA FORMAT

D.1 SCOPE

D.1.1 Scope. The one-time DID will typically need additional format definitions to support the program specific data requirements. A sample supplemental formatting matrix is included in D.2.1.

D.2 SAMPLE TDP DRAWINGS METADATA FORMAT (AIRCRAFT CARRIER EXAMPLE)

D.2.1 TDP Drawings Metadata Format. An example of the TDP drawings metadata format can be seen in table D-1.

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TABLE D-1. TDP drawings metadata format matrix example.

Field Name	Business Definition	Alternative Names	Unique Value	Data Type	Length	Valid Values	Default Value	Calculation	Required	Business Rule	User Role	Owner Role	Admin Role
Drawing Number	The number used to uniquely identify a TDP element	DWG No	Yes	Alpha numeric	32	May contain only "-" (dash) special character, alpha characters, or numbers	Null	N/A	Yes	Must be in accordance with ASME Y14.100-2013, Section 6	View	View, Edit	View, Edit, Delete
Drawing Revision	The current revision of a document, drawing, or list	DWG Rev	No	Alpha numeric	3	May contain only "-" (dash) special character or alpha characters	Null	N/A	Yes	The revision assigned to the drawing record must be in accordance with ASME Y14.35	View	View, Edit	View, Edit, Delete
currentrev	Is this the Current Rev?			Alpha	1	Y,N	Null	N/A	Yes		View		
Drawing Type	The type of drawing, document, or list	DWG Type	No	Alpha numeric	50	May contain only alpha characters, numbers, and special characters	Null	N/A	Yes	Must be a valid drawing type in accordance with ASME Y14.24-2012, Table A-1	View	View, Edit	View, Edit, Delete
Title	The title of the drawing, document, or list	Nomenclature	No	Alpha numeric	99	May contain only alpha characters and numbers	Null	N/A	Yes	Must be in accordance with ASME Y14.100-2013, Section 5	View	View, Edit	View, Edit, Delete

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TABLE D-1. TDP drawings metadata format matrix example – Continued.

Field Name	Business Definition	Alternative Names	Unique Value	Data Type	Length	Valid Values	Default Value	Calculation	Required	Business Rule	User Role	Owner Role	Admin Role
Drawing CAGE	The CAGE code identified on the drawing, document, or list (Design Activity Identifier)	DAI	No	Alpha numeric	5	May contain only alpha characters and numbers	Null	N/A	Yes	Must be a valid CAGE	View	View, Edit	View, Edit, Delete
System	The system the item is used on	N/A	No	Alpha numeric	32	May contain only "-" (dash) special character, alpha characters, or numbers	Null	N/A	Yes	Must be a valid System	View	View, Edit	View, Edit, Delete
Release Date	The date the item was authorized for release	N/A	No	Date	12	DD-MMM-YYYY	Null	N/A	No	If left empty the date imported will be assigned as the release date	View	View, Edit	View, Edit, Delete
Contract	Contract listed on the drawing, document, or list	N/A	No	Alpha numeric	25	May contain only alpha characters and numbers	Null	N/A	Yes	Contract under which the item was developed and the data was created	View	View, Edit	View, Edit, Delete

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TABLE D-1. TDP drawings metadata format matrix example – Continued.

Field Name	Business Definition	Alternative Names	Unique Value	Data Type	Length	Valid Values	Default Value	Calculation	Required	Business Rule	User Role	Owner Role	Admin Role
Security	A value to denote the security classification value	N/A	No	Alpha numeric	1	May contain only alpha characters	Null	N/A	Yes	Valid Values are UNCLASSIFIED (U), CONFIDENTIAL (C), SECRET (S), or TOP SECRET (T)	View	View, Edit	View, Edit, Delete
Next Higher Drawing	The next higher assembly	NHA	No	Alpha numeric	32	May contain only "-" (dash) special character, alpha characters, or numbers	Null	N/A	Yes	Must be a valid drawing number	View	View, Edit	View, Edit, Delete
Next Higher Drawing Type	Drawing type of the next higher assembly	NHA DT	No	Alpha numeric	20	May contain only alpha characters, numbers, and special characters	Null	N/A	Yes	Must be a valid drawing type in accordance with ASME Y14.24-2012, Table A-1	View	View, Edit	View, Edit, Delete
Distribution Statement	Distribution statement of the drawing/ document	N/A	No	Alpha	1	A, B, C, D, E, F, X	no default	N/A	Yes	Must be in accordance with DoDI 5230.24 and NAVAIRINST 5230.24	View	View, Edit	View, Edit, Delete
Approval Category													

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D.2.2 TDP parts metadata format. An example of the TDP parts metadata format can be seen in table D-2.

TABLE D-2. TDP parts metadata format matrix example.

Field Name	Business Definition	Alternative Names	Unique Value	Data Type	Length	Valid Values	Default Value	Calculation	Required	Business Rule	User Role	Owner Role	Admin Role
Drawing Number	The number used to uniquely identify a TDP element	DWG No	Yes	Alpha numeric	32	May contain only "-" (dash) special character, alpha characters, or numbers	Null	N/A	Yes	Must be in accordance with ASME Y14.100-2013, Section 6	View	View, Edit	View, Edit, Delete
Drawing Revision	The current revision of a document, drawing, or list	DWG Rev	No	Alpha numeric	3	May contain only "-" (dash) special character or alpha characters	Null	N/A	Yes	The revision assigned to the drawing record must be in accordance with ASME Y14.35	View	View, Edit	View, Edit, Delete
WBS	WBS breakdown	N/A	Yes	Alpha numeric	25	May contain only alpha characters and numbers	Null	N/A	No	Required for new acquisitions. Must be in accordance with MIL-STD-881	View	View, Edit	View, Edit, Delete
Part Identifier													

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TABLE D-2. TDP parts metadata format matrix example – Continued.

Field Name	Business Definition	Alternative Names	Unique Value	Data Type	Length	Valid Values	Default Value	Calculation	Required	Business Rule	User Role	Owner Role	Admin Role
CAGE	The CAGE code identified on the drawing, document, or list (Design Activity Identifier)	DAI	No	Alpha numeric	5	May contain only alpha characters and numbers	Null	N/A	Yes	Must be a valid CAGE	View	View, Edit	View, Edit, Delete
Ref Des	Reference designator of an item in a system	Reference Designator	Yes	Alpha numeric	12	May contain only alpha characters and numbers	Null	N/A	No	Must be a reference designation in accordance with ASME Y14.44	View	View, Edit	View, Edit, Delete
Part No	Part number(s) of item being defined by the drawing, model, or list	PN, Reference No., Ref No.	No	Alpha numeric	32	May contain only alpha characters, special characters, and numbers	Null	N/A	Yes	Part number of item being defined in drawing, model, or list. If multiple part numbers are defined, then a single entry must be made for each part number	View	View, Edit	View, Edit, Delete
Part Revision	If part revisioning is used, then indicate revision associated with part	N/A	No	Alpha numeric	15	May contain only alpha characters and numbers	Null	N/A	No	Only required when a revision is part of the part identification number	View	View, Edit	View, Edit, Delete

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TABLE D-2. TDP parts metadata format matrix example – Continued.

Field Name	Business Definition	Alternative Names	Unique Value	Data Type	Length	Valid Values	Default Value	Calculation	Required	Business Rule	User Role	Owner Role	Admin Role
Part CAGE	CAGE code for the source of supply for the part. This may not be the same as the Design Activity CAGE or DAI	Vendor CAGE	No	Alpha numeric	5	May contain only alpha characters and numbers	Null	N/A	Yes	Must be a valid CAGE	View	View, Edit	View, Edit, Delete
Description	Item name or nomenclature (i.e., screw, bolt, cable transmitter)	N/A	No	Alpha numeric	99	May contain only alpha characters and numbers	Null	N/A	Yes	Must be a common noun name, title, or nomenclature	View	View, Edit	View, Edit, Delete

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TABLE D-2. TDP parts metadata format matrix example – Continued.

Field Name	Business Definition	Alternative Names	Unique Value	Data Type	Length	Valid Values	Default Value	Calculation	Required	Business Rule	User Role	Owner Role	Admin Role
Data Rights	Data Rights category asserted by data developer	DR	No	Alpha	4	May select from the following values: "LR" – Limited Rights; "GPR" – Govt Purpose Rights; "SNLR" – Specially Negotiated License Rights; "UL" – Unlimited Rights; "COTS" – Commercial Off the Shelf; "PGR" – Prior Govt Rights; "SIBR" – SBIR data rights	Null	N/A	Yes	Select from the following: LR, GPR, SNLR, UL, COTS, PGR, or SBIR	View	View, Edit	View, Edit, Delete
Shock Grade													
Part Approvals													
Hazmat Identification													

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D.2.3 TDP files metadata format. An example of the TDP files metadata format can be seen on table D-3.

TABLE D-3. TDP files metadata format matrix example.

Field Name	Business Definition	Alternative Names	Unique Value	Data Type	Length	Valid Values	Default Value	Calculation	Required	Business Rule	User Role	Owner Role	Admin Role
Drawing Number	The number used to uniquely identify a TDP element	DWG No	Yes	Alpha numeric	32	May contain only "-" (dash) special character, alpha characters, or numbers	Null	N/A	Yes	Must be in accordance with ASME Y14.100-2013, Section 6	View	View, Edit	View, Edit, Delete
Drawing Revision	The current revision of a document, drawing, or list	DWG Rev	No	Alpha numeric	3	May contain only "-" (dash) special character or alpha characters	Null	N/A	Yes	The revision assigned to the drawing record must be in accordance with ASME Y14.35	View	View, Edit	View, Edit, Delete
CAGE	The CAGE code identified on the drawing, document, or list (Design Activity Identifier)	DAI	No	Alpha numeric	5	May contain only alpha characters and numbers	Null	N/A	Yes	Must be a valid CAGE	View	View, Edit	View, Edit, Delete

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TABLE D-3. TDP files metadata format matrix example – Continued.

Field Name	Business Definition	Alternative Names	Unique Value	Data Type	Length	Valid Values	Default Value	Calculation	Required	Business Rule	User Role	Owner Role	Admin Role
Filename	Drawing, model, list, or document filenames(s). Complete filenames(s) with file extension.	N/A	Yes	Alpha numeric	99	May contain alpha characters, special characters, and numbers	Null	N/A	Yes	Filename for data file of the drawing, model, associated list, or document. Filename for native file defining the item with file extension. If the item is an assembly requiring multiple files to be loaded in the native software, then all files must be zipped (compressed) into a single file folder using the following naming convention: "CAGE" "drawing number" "Rev" "revision letter(s)" with the ".zip" file extension.	View	View, Edit	View, Edit, Delete

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TABLE D-3. TDP files metadata format matrix example – Continued.

Field Name	Business Definition	Alternative Names	Unique Value	Data Type	Length	Valid Values	Default Value	Calculation	Required	Business Rule	User Role	Owner Role	Admin Role
Frame Number	The number of the frame	N/A	No	Numeric	4	May contain only numbers	Null	N/A	No	When drawing or list is cut into multiple frames, this should be used to identify the frame	View	View, Edit	View, Edit, Delete
Total Frames	Total count of frames	N/A	No	Numeric	4	May contain only numbers	Null	N/A	No	When drawing or list is cut into multiple frames, this should be used to identify the frame total	View	View, Edit	View, Edit, Delete
Sheet Number	The number of the sheet	N/A	No	Numeric	4	May contain only numbers	Null	N/A	No	When drawing or list is cut into multiple sheets, this should be used to identify the sheet	View	View, Edit	View, Edit, Delete
Total Sheets	Total count of sheets	N/A	No	Numeric	4	May contain only numbers	Null	N/A	No	When drawing or list is cut into multiple sheets, this should be used to identify the sheet total	View	View, Edit	View, Edit, Delete

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TABLE D-3. TDP files metadata format matrix example – Continued.

Field Name	Business Definition	Alternative Names	Unique Value	Data Type	Length	Valid Values	Default Value	Calculation	Required	Business Rule	User Role	Owner Role	Admin Role
Sheet Revision	Revision of the sheet	N/A	No	Alpha	4	May contain only "-" (dash) special character or alpha characters	Null	N/A	No	When drawing or list is cut into multiple sheets, this should be used to identify the sheet revision	View	View, Edit	View, Edit, Delete

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APPENDIX E. SAMPLE FUNCTIONAL GROUPING OF DRAWINGS

E.1 SCOPE

E.1.1 Scope. A sample supplemental formatting matrix to define drawing types is included in E.2 to support the one-time DID.

E.2 SAMPLE FUNCTIONAL GROUPING OF DRAWINGS

Functional Grouping of Drawings	Drawing Type	Abbreviation	Business Rule
Control drawings	Procurement control drawing	PCD	IAW ASME Y14.24-2012, Table A-1
	Vendor item control drawing	VICD	IAW ASME Y14.24-2012, Table A-1
	Source control drawing	SCD	IAW ASME Y14.24-2012, Table A-1
	Envelope drawing	EVD	IAW ASME Y14.24-2012, Table A-1
	Identification cross-reference drawing	ICRD	IAW ASME Y14.24-2012, Table A-1
	Microcircuit drawing set	MDS	IAW ASME Y14.24-2012, Table A-1
Design information drawings	Layout drawing	LOD	IAW ASME Y14.24-2012, Table A-1
	Installation drawing	IND	IAW ASME Y14.24-2012, Table A-1
	Arrangement drawing	ARD	IAW ASME Y14.24-2012, Table A-1
	Interface drawing	IFD	IAW ASME Y14.24-2012, Table A-1
	Mechanical schematic diagram	MSD	IAW ASME Y14.24-2012, Table A-1
	Functional block diagram	FBD	IAW ASME Y14.24-2012, Table A-1
	Single-line diagram	SLD	IAW ASME Y14.24-2012, Table A-1
	Schematic or circuit diagram	SD	IAW ASME Y14.24-2012, Table A-1
	Connection diagram	CD	IAW ASME Y14.24-2012, Table A-1
	Wiring diagram	WD	IAW ASME Y14.24-2012, Table A-1
	Interconnection diagram	ICD	IAW ASME Y14.24-2012, Table A-1
	Wiring list	WL	IAW ASME Y14.24-2012, Table A-1
	Logic circuit diagram	LCD	IAW ASME Y14.24-2012, Table A-1
	Drawing tree	DT	IAW ASME Y14.24-2012, Table A-1
Hardware and software drawings	Layout drawing	LOD	IAW ASME Y14.24-2012, Table A-1
	Detail drawing	DD	IAW ASME Y14.24-2012, Table A-1
	Monodetail drawing	MOND	IAW ASME Y14.24-2012, Table A-1
	Multidetail drawing	MUDD	IAW ASME Y14.24-2012, Table A-1
	Master drawing	MD	IAW ASME Y14.24-2012, Table A-1
	Tube bend drawing	TUBD	IAW ASME Y14.24-2012, Table A-1
	Contour definition drawing	CDD	IAW ASME Y14.24-2012, Table A-1
	Software installation drawing	SOID	IAW ASME Y14.24-2012, Table A-1
	Assembly drawing	ASSY	IAW ASME Y14.24-2012, Table A-1
	Inseparable assembly drawing	IAD	IAW ASME Y14.24-2012, Table A-1
	Wiring harness drawing	WHD	IAW ASME Y14.24-2012, Table A-1
	Cable assembly drawing	CAD	IAW ASME Y14.24-2012, Table A-1

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Board assembly drawing	BAD	IAW ASME Y14.24-2012, Table A-1
Tube bend drawing	TUBD	IAW ASME Y14.24-2012, Table A-1
Installation drawing	IND	IAW ASME Y14.24-2012, Table A-1
Altered item drawing	AID	IAW ASME Y14.24-2012, Table A-1
Selected item drawing	SID	IAW ASME Y14.24-2012, Table A-1
Modification drawing	MODD	IAW ASME Y14.24-2012, Table A-1
Wiring list	WL	IAW ASME Y14.24-2012, Table A-1
Artwork	ARW	IAW ASME Y14.24-2012, Table A-1
Artwork master	ARWM	IAW ASME Y14.24-2012, Table A-1
Production master	PRODM	IAW ASME Y14.24-2012, Table A-1
Master pattern drawing	MPD	IAW ASME Y14.24-2012, Table A-1
Undimensioned drawing	UDD	IAW ASME Y14.24-2012, Table A-1
Kit drawing	KD	IAW ASME Y14.24-2012, Table A-1
Matched set drawing	MSTD	IAW ASME Y14.24-2012, Table A-1
Specification	SPEC	
Standard	STD	

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APPENDIX F. SAMPLE VALUES FOR ASSOCIATED LISTS

F.1 SCOPE

F.1.1 Scope. A sample supplemental formatting matrix to define data types is included in F.2, to support the one-time DID, as specified in ASME Y14.34.

F.2 SAMPLE VALUES FOR ASSOCIATED LISTS

List Type	Prefix (Abbreviation)	Business Rule
Parts List	PL	IAW ASME Y14.34
Application List	AL	IAW ASME Y14.34
Data List	DL	IAW ASME Y14.34
Index List	IL	IAW ASME Y14.34
Wire List	WL	IAW ASME Y14.34

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APPENDIX G. SAMPLE VALUES FOR DATA RIGHTS CATEGORIES

G.1 SCOPE

G.1.1 Scope. A sample supplemental formatting matrix to define data rights categories is included in G.2, to support the one-time DID.

G.2 SAMPLE VALUES FOR DATA RIGHTS CATEGORIES

Data Rights Category	Abbreviation (Acronym)
Limited Rights	LR
Government Purpose Rights	GPR
Unlimited Rights	UL
Special Negotiated License Rights	SNLR
Commercial Off The Shelf	COTS
Prior Government Rights	PGR
Small Business Innovation Research	SBIR

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APPENDIX H. ADDITIONAL STANDARDS FOR DATA EXCHANGE

H.1 SCOPE

H.1.1 Scope. H.2 provides a list of additional references that capture the various data exchange standards.

H.2 APPLICABLE DOCUMENTS

ECMA INTERNATIONAL

ECMA-363 - Universal 3D File Format (U3D)

ECMA-376 - Office Open XML File Formats

(Copies of these documents are available online at https://www.ecma-international.org/.)

INTERNATIONAL COMMITTEE FOR INFORMATION TECHNOLOGY STANDARDS (INCITS)

INCITS/ISO/IEC 26300 - Open Document Format for Office Applications (OpenDocument)

(Copies of this document are available online at https://standards.incits.org/.)

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO/IEC 19775 - Information Technology - Computer Graphics, Image Processing and Environmental Data Representation - Extensible 3D (X3D)

ISO 24517-1 - Document Management - Engineering Document Format using PDF

ISO/IEC 29500 - Information Technology - Document Description and Processing Languages - Office Open XML File Formats

ISO 32000 - Document Management - Portable Document Format (PDF)

(Copies of these documents are available online at www.iso.org.)

NATIONAL INFORMATION STANDARDS ORGANIZATION (NISO)

ANSI/NISO Z39.18 - Scientific and Technical Reports - Preparation, Presentation, and Preservation

(Copies of this document are available online at https://www.niso.org/.)

U.S. PRODUCT DATA ASSOCIATION (US PRO)

Initial Graphics Exchange Specification (IGES) 5.3

(Copies of this document are available online at https://infostore.saiglobal.com.)

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APPENDIX J. PRODUCT MODEL SAMPLE VIEWS

J.1 SCOPE

J.1.1 Benefits of 3D product models. 3D product data provides many potential benefits over 2D product data. The 3D model is used in many applications within the engineering community to drive efficiencies. Some of the benefits include:

a. Provides enhanced clarity on product drawings
b. Supports training requirements and publications
c. Simplifies and improve effectiveness of technical manuals
d. Supports shipyard and depot electronic maintenance, repairs, and process illustrations
e. Improves communication within technical briefs

Examples of various 3D product models are included in this appendix.

J.2. 3D PRODUCT MODEL SAMPLE VIEWS

J.2.2 Rotated assembled view 3D product model. An example of a rotated assembled view 3D product model can be seen on figure J-1.

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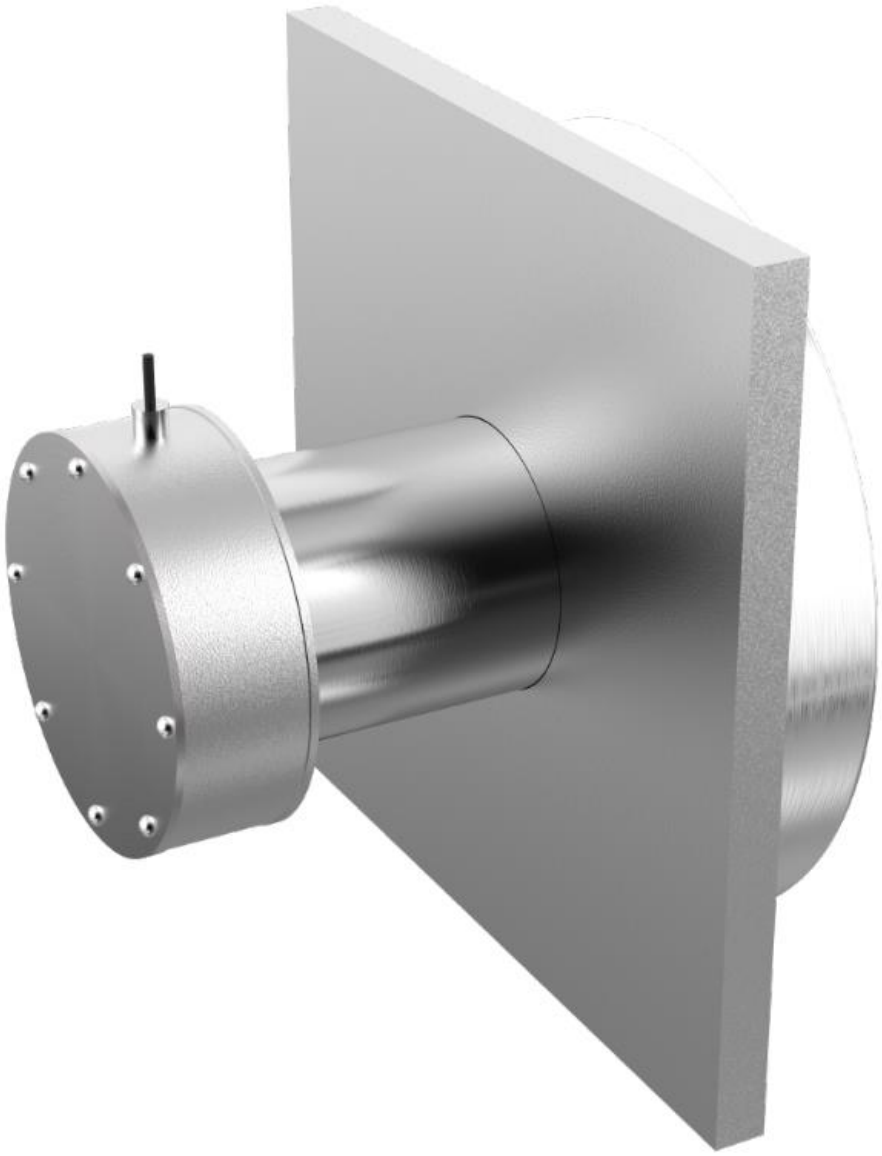


FIGURE J-1. Example of a rotated assembled view 3D model.

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J.2.3 Exploded view 3D product model. An example of an exploded view 3D product model can be seen on figure J-2.

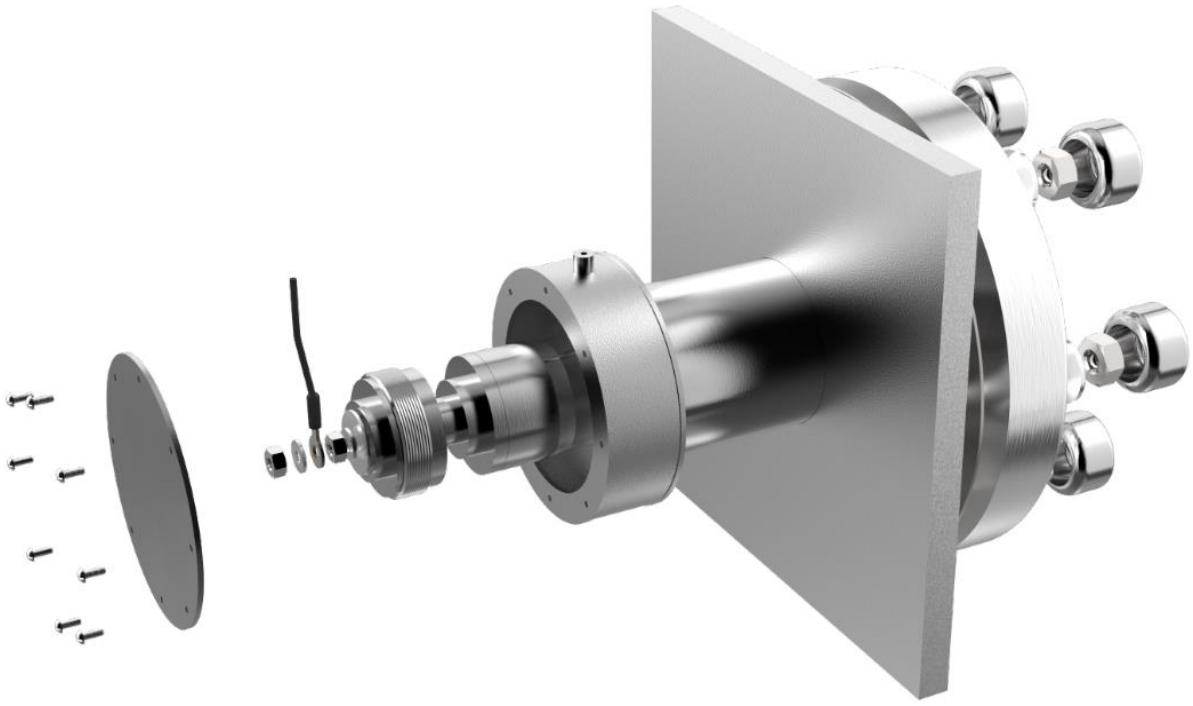


FIGURE J-2. Example of an exploded view 3D product model.

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J.2.4 Glassed view 3D product model. An example of a glassed view 3D product model can be seen on figure J-3.

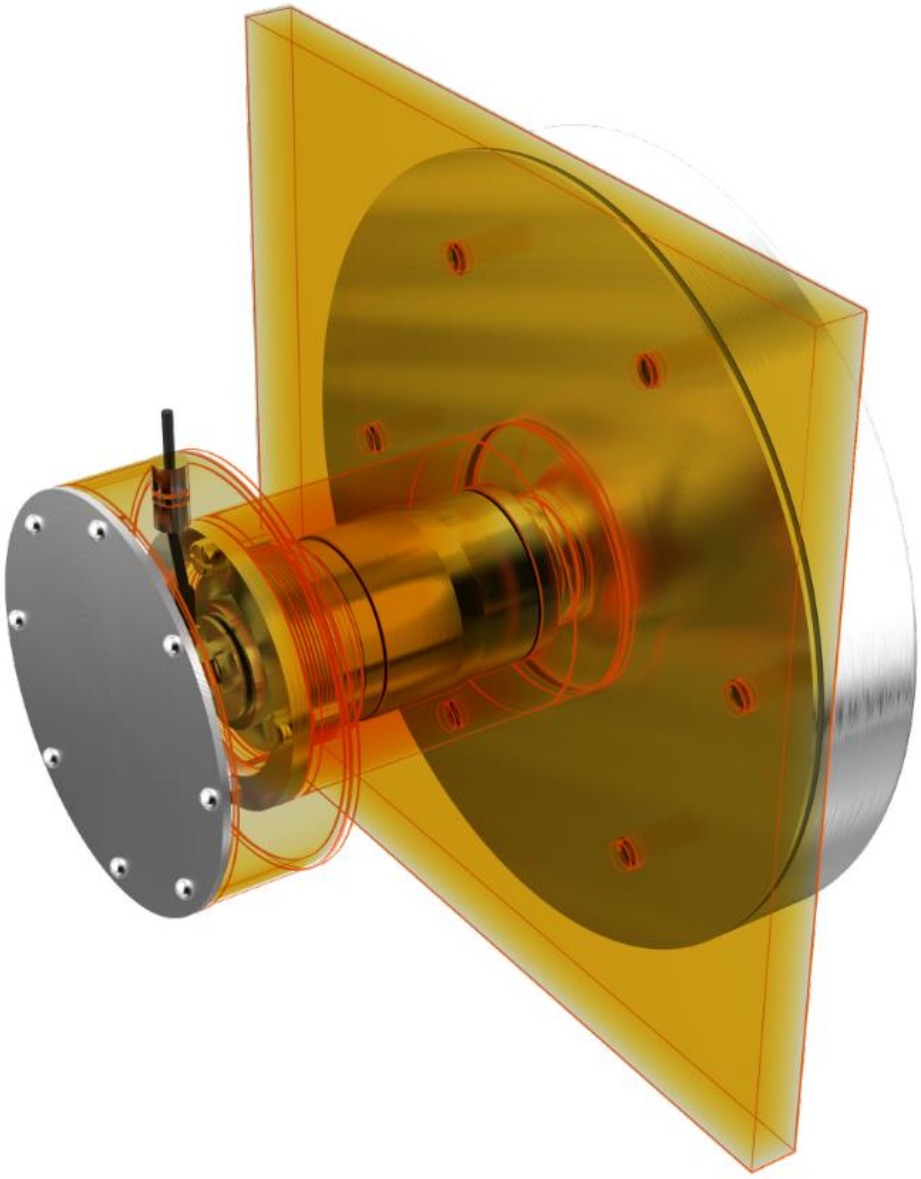


FIGURE J-3. Example of a glassed view 3D product model.

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J.2.5 Sectioned view with color 3D product model. An example of a sectioned view with color 3D product model can be seen on figure J-4.

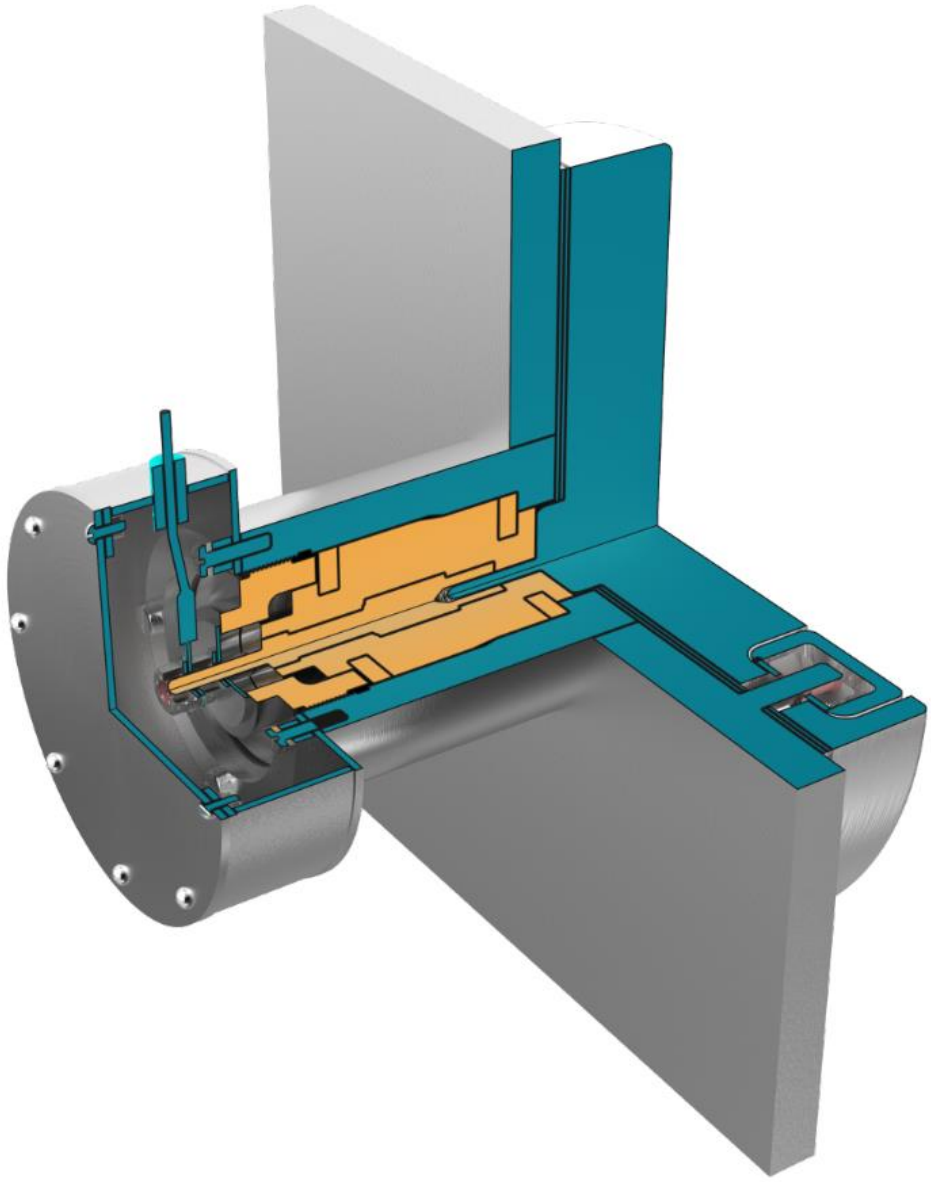


FIGURE J-4. Example of a sectioned view with color 3D product model.

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CONCLUDING MATERIAL

Custodians:
- Army – MI
- Navy – SH
- Air Force – 01
- DLA – DH
- OSD – SE

Preparing activity:
- Navy – SH
- (Project SESS-2022-011)

Review activities:
- Army – AC, AR, AV, CR, PT, SY, TE, TM
- Navy – AS, CG, CH, EC, MC, OS, SA, NM
- Air Force – 08, 10, 11, 16, 19, 24, 33, 70, 84, 94
- DLA – CC, CQ, CT, GS
- DISA – DC1, DC5
- MDA – MDA
- NSA – NS
- OSD – DMS, MA, RS, SO
- USTC – USTC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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Check the source to verify that this is the current version before use.